“APNA MART ONLINE GROCERY STORE”

A

Project Report

Submitted in partial fulfillment of the requirement for the

award ofDegree of

Bachelor of Computer Applications

Submitted to

452 – Cimage College, Patna

Submitted By
SATYAM KUMAR
Registration No.**202345200077**
BCA-3rd Year
Session-2023-2026

Under Joint Guidance of
Prof. Raju Upadhyay &
Prof. Sanjeev Kumar
[Asst. Professor]
Computer Science &
Applications

Cimage College, Patna

Department of Computer Science & Applications

Session 2023-2026

DATE:……

CERTIFICATE

This is to certify that the work embodies in this project entitled, **"APNA MART ONLINE GROCERY STORE",** being submitted by **SATYAM KUMAR (202345200077)** in partial fulfillment of the requirement for the award of "**Bachelor of Computer Applications**" to ***PATLIPUTRA UNIVERSITY, PATNA*** during the academic year 2022-25 is a record of bonafide piece of work, carried out by him/her under our/my supervision and guidance in the "**Department of Computer Science & Applications**", **Cimage College, Patna.**

Dr. Neeraj Agrawal	**Dr. Amit Kumar Shukla**	**Prof. Nitish Kumar Rohatgi**
Director	Head of Department	Principal

Date:.....

APPROVAL CERTIFICATE

The project report entitled **"APNA MART ONLINE GROCERY STORE",** being submitted by SATYAM KUMAR(**202345200077**) has been examined by us and is hereby approved for the award of degree "**Bachelor of Computer Applications",** for which it has been submitted. It is understood that by this approval the undersigned do not necessarily endorse or approve any statement made, the opinion expressed or conclusion drawn therein, but approve the project only for the purpose for which it has been submitted.

(Internal Examiner) (External Examiner)

Date: Date:

Date:.....

DECLARATION

I, **SATYAM KUMAR** hereby declare that the work, which is being presented in the project report entitled **"APNA MART ONLINE GROCERY STORE",** partial fulfillment of the requirements for the award of degree of "**Bachelor of Computer Applications**" submitted in the department of **Computer Science & Applications (Cimage College)** is an authentic record of my own work carried under the joint guidance of **Prof. Raju Upadhyay Sir and Prof. Sanjeev Kumar Sir.** I have not submitted the matter embodied in this report for the award of any other degree.

SAYTAM KUMAR

Registration No:- 202345200077

BCA 3rd Year

Session-2023-2026

ACKNOWLEDGEMENT

"A journey is easier when you travel together. Interdependence is certainly more valuable than independence."

I would like to thanks, **Prof. Raju Upadhyay Sir** and **Prof. Sanjeev Kumar Sir**, for providing regular guidance and insight into my project work. I also thank them for all advice he has givenme in the past year, and for always having time for me, whenever I needed.

I give special thanks to **Dr. Amit Kumar Shukla Sir, Prof. Ravi Kumar Soni Sir,** and **Prof. Anjesh Kumar Sir** for always being willing to help find solutions to any problems I had with my work.

"The completion of any project depends upon the cooperation, coordination, and combined efforts of several resources of knowledge, inspiration, and energy".

I also extend my deepest gratitude to **Director**, CIMAGE GROUP OF INSTITUITIONS, PATNA, **Dr. Neeraj Agrawal Sir, Dean** CIMAGE GROUP OF INSTITUITIONS, PATNA, **Dr. Neeraj Kumar Sir** and **Centre Head Mrs. Megha Agrawal Ma'am** for providing all the necessary facilities and true encouraging environment to bring out the best in my endeavors.

I express my gratitude and thanks to all the staff members of Computer Science departments for their sincere cooperation in furnishing relevant information to complete this Project well in time successfully.

I extend a special word to my friends, who have been a constant source of inspiration throughout my project work.

Lastly but not least I must express my cordial thank to my parent and family members who gave me the moral support without that it is impossible to complete my project work. With this note, I thank everyone for the support.

SAYTAM KUMAR

Registration No:- 202345200077

BCA 3rd Year

Session-2023-2026

Index table: Page No

Introduction..... 1

Objective..... 1 – 2

Project Overview..... 2

Feasibility Study..... 2 – 4

System Analysis..... 4 – 5

System Specifications & Design ..... 5 – 6

Project Planning & Scheduling..... 6

Gantt Chart..... 6 – 7

Pert Chart..... 7 – 8

DFD..... 9 – 10

Module Description..... 11 – 12

Data Dictionary.....12 – 14

ER Diagram.....14 – 16

Scope and Future Enhancement..... 17

Implementation Methodology..... 17 – 18

Testing..... 19 – 20

Limitations and Further Enhancements..... 20 – 21

Security Mechanism..... 21 – 22

Database Model..... 22 – 23

Files of the project in Image..... 24

Teacher Module Code & Snapshots..... 23 – 70

Student Module Code & Snapshots..... 70 -129

Conclusion..... 130

Bibliography..... 131 – 139

INTRODUCTION (Online Grocery Store – ApnaMart)

The rapid growth of information technology and the internet has significantly transformed the way people perform their daily activities, including shopping. In recent years, online shopping has become an integral part of modern life due to its convenience, accessibility, and time-saving nature. Among various categories of online shopping, grocery shopping has gained tremendous popularity as people increasingly prefer purchasing essential household items from the comfort of their homes. The Online Grocery Store system, titled "ApnaMart," is developed to meet this growing demand by providing a simple, efficient, and user-friendly platform for customers to buy groceries online.

Traditionally, grocery shopping involves visiting physical stores, selecting items manually, waiting in long queues, and carrying heavy bags back home. This process is not only time-consuming but also inconvenient, especially for working professionals, elderly individuals, and people with busy lifestyles. Moreover, situations like bad weather, traffic congestion, or health concerns (such as pandemics) further highlight the need for an alternative shopping solution. The Online Grocery Store system addresses these challenges by offering a digital platform where users can browse products, compare prices, and place orders at any time without leaving their homes.

The primary aim of the ApnaMart system is to simplify the grocery shopping process and make it more efficient. The system provides a wide range of products, including fruits, vegetables, dairy items, beverages, and packaged goods, organized into categories for easy navigation. Users can create accounts, log in securely, and manage their shopping activities. The intuitive user interface ensures that even individuals with minimal technical knowledge can easily use the system. With just a few clicks, customers can add items to their cart, review their selections, and proceed to checkout.

One of the key features of the Online Grocery Store is the shopping cart functionality, which allows users to select multiple products and manage their purchases before placing an order. This feature mimics the real-life shopping experience while offering additional flexibility. Users can update quantities, remove items, and view the total cost of their purchase in real time. The checkout process collects essential information such as delivery address, contact details, and payment method, ensuring that orders are processed efficiently.

In addition to the user module, the system includes an admin module that plays a crucial role in managing the entire application. The admin panel allows administrators to add, update, or delete products, manage categories, and

🎯 OBJECTIVE (Online Grocery Store – ApnaMart)

The primary objective of the Online Grocery Store system, titled "ApnaMart," is to design and develop a web-based application that simplifies the process of purchasing grocery items by providing a convenient, efficient, and user-friendly platform for customers. With the rapid advancement of technology and the increasing use of the internet in daily life, there is a growing demand for online services that can reduce manual effort and save time. This project aims to fulfill that demand by creating a system that enables users to shop for groceries online from anywhere and at any time.

One of the main objectives of this project is to provide a seamless and hassle-free shopping experience to users. Traditional grocery shopping requires customers to visit physical stores, search for products, stand in queues, and carry items back home. This process can be tiring and time-consuming, especially for working individuals, elderly people, and those living in urban areas with heavy traffic. The Online Grocery Store system eliminates these difficulties by allowing users to browse products, select items, and place orders with just a few clicks. This enhances convenience and improves the overall shopping experience.

Another important objective is to develop a system that is easy to use and accessible to all types of users, including those with limited technical knowledge. The user interface of the application is designed to be simple, intuitive, and interactive. Clear navigation menus, categorized product listings, and search functionality help users find products quickly and efficiently. The system ensures that even first-time users can easily understand how to use the platform without requiring any technical training.

The project also aims to provide efficient product management and organization. In a physical store, managing a large number of products can be challenging. However, in an online system, products can be categorized, sorted, and displayed in an organized manner. This project includes features such as product categories, product descriptions, pricing details, and images, which help users make informed purchasing decisions. The objective is to create a well-structured system that allows easy addition, modification, and deletion of products by the administrator.

A key objective of the ApnaMart system is to implement a reliable and efficient cart management system. The shopping cart is an essential feature of any e-commerce platform, as it allows users to select multiple items before making a purchase. This project provides a dynamic cart system where users can add

📊 PROJECT OVERVIEW (Online Grocery Store – ApnaMart)

The Online Grocery Store system, named "ApnaMart," is a web-based application designed to provide an efficient and user-friendly platform for purchasing grocery items online. This project focuses on automating the traditional grocery shopping process by allowing users to browse products, add items to a cart, and place orders from the comfort of their homes. The system integrates both frontend and backend technologies to ensure smooth operation, efficient data management, and a seamless user experience.

In today's fast-paced world, people prefer solutions that save time and effort. Traditional grocery shopping involves visiting physical stores, searching for products, waiting in long queues, and carrying items back home. These challenges can be inconvenient, especially for working professionals, elderly individuals, and people living in busy urban areas. The ApnaMart system addresses these issues by providing an online platform that allows users to shop anytime and from anywhere. This convenience makes online grocery shopping an increasingly popular choice among consumers.

The project is designed using a client-server architecture, where the frontend serves as the user interface and the backend handles data processing and storage. The frontend of the application is developed using HTML, CSS, and JavaScript, which provide an interactive and visually appealing interface. The backend is implemented using PHP, which processes user requests, handles authentication, and interacts with the database. MySQL is used as the database management system to store data related to users, products, orders, and transactions.

The system is divided into two main components: the user module and the admin module. These modules work together to ensure the proper functioning of the application. The user module focuses on providing customers with an easy and convenient shopping experience, while the admin module allows administrators to manage the system efficiently.

The user module is the primary interface through which customers interact with the system. It includes features such as user registration, login, product browsing, cart management, and order placement. Users can create an account by providing basic information such as name, email, and password. Once registered, they can log in to the system and access various features. The login system ensures that only authorized users can access their accounts and perform transactions.

📈 FEASIBILITY STUDY (Online Grocery Store)

Feasibility study is an important step in the development of any system. It is conducted to determine whether the proposed system is practical, beneficial, and capable of being implemented successfully. The feasibility study helps in analyzing different aspects of the project such as technical requirements, cost, operational efficiency, and overall benefits. For the Online Grocery Store system (ApnaMart), the feasibility study ensures that the system can be developed using available resources and can meet the needs of users effectively.

The feasibility study of this project is divided into several key components, including technical feasibility, economic feasibility, operational feasibility, legal feasibility, and schedule feasibility. Each of these aspects is analyzed in detail to evaluate the practicality of the system.

🔷 1. TECHNICAL FEASIBILITY

Technical feasibility refers to the ability of the proposed system to be developed and implemented using existing technology and resources. It evaluates whether the required hardware, software, and technical expertise are available.

The Online Grocery Store system is developed using widely used and easily available technologies such as HTML, CSS, and JavaScript for the frontend, PHP for the backend, and MySQL for the database. These technologies are well-supported, reliable, and suitable for web-based applications. They are also easy to learn and implement, making them ideal for academic projects.

The hardware requirements for the system are minimal. A basic computer or laptop with moderate specifications is sufficient to develop and run the application. Additionally, an internet connection is required for accessing and testing the system. The use of XAMPP server allows the system to be run locally, eliminating the need for expensive hosting services during development.

The system is designed using a modular approach, which makes it easy to develop, test, and maintain. Each module, such as user management, product management, and order processing, is developed separately and then integrated into the main system. This approach reduces complexity and improves reliability.

Another important factor in technical feasibility is system performance. The Online Grocery Store system is designed to handle multiple users and transactions efficiently. The use of a relational database management system

(MySQL) ensures fast data retrieval and storage. Proper indexing and query optimization techniques can further enhance performance.

Security is also considered in technical feasibility. The system includes basic security features such as user authentication, password protection, and input validation. These measures help in protecting user data and preventing unauthorized access.

Overall, the Online Grocery Store system is technically feasible as it can be developed using available technologies, tools, and resources without any major challenges.

◆ 2. ECONOMIC FEASIBILITY

Economic feasibility evaluates the cost-effectiveness of the proposed system. It determines whether the benefits of the system outweigh the costs involved in its development and implementation.

The Online Grocery Store system is highly economical as it is developed using free and open-source tools. Software such as XAMPP, Visual Studio Code, and MySQL are available at no cost. This significantly reduces the overall development cost.

The hardware requirements are also minimal, as the system can be developed and tested on a standard computer or laptop. There is no need for expensive infrastructure or specialized equipment. Additionally, the system can be hosted on low-cost or free hosting platforms if required.

The benefits of the system are significant. It reduces manual effort, saves time, and improves efficiency. For businesses, it provides an opportunity to reach a larger customer base and increase sales. It also reduces operational costs associated with maintaining physical stores, such as rent, electricity, and staff salaries.

The system also helps in better inventory management and sales tracking, which can lead to improved decision-making and increased profitability. By automating various processes, the system reduces the chances of errors and improves accuracy.

From a long-term perspective, the system offers high returns on investment. Once developed, it can be used for an extended period with minimal maintenance costs. Future enhancements can be added without significant additional investment.

Therefore, the Online Grocery Store system is economically feasible as it requires low investment and provides high benefits.

🔷 3. OPERATIONAL FEASIBILITY

Operational feasibility evaluates how well the proposed system will work in the real-world environment and whether it will be accepted by users.

The Online Grocery Store system is designed to be user-friendly and easy to operate. The interface is simple, intuitive, and easy to navigate. Users can easily register, log in, browse products, and place orders without any difficulty. This ensures that the system can be used by people with different levels of technical knowledge.

The system also provides flexibility, as users can access it from anywhere and at any time. This makes it convenient for users and increases the likelihood of acceptance. The 24/7 availability of the system ensures that users are not restricted by store timings.

The admin module is also designed to be easy to use. Administrators can manage products, orders, and users through a simple dashboard. This reduces the need for extensive training and makes the system easy to maintain.

The system improves efficiency by automating various processes such as product management, order processing, and data storage. This reduces manual effort and minimizes errors. It also ensures faster processing of orders and better customer service.

User satisfaction is a key factor in operational feasibility. The system provides features such as search functionality, product categorization, and real-time updates, which enhance the user experience. These features make the system more efficient and user-friendly.

Overall, the Online Grocery Store system is operationally feasible as it is easy to use, efficient, and likely to be accepted by users.

🔷 4. LEGAL FEASIBILITY

Legal feasibility refers to whether the proposed system complies with legal requirements and regulations.

The Online Grocery Store system does not violate any legal norms, as it is developed for academic purposes. However, if the system is implemented in a real-world scenario, certain legal aspects need to be considered.

These include:

- Data protection and privacy laws
- Secure handling of user information
- Compliance with e-commerce regulations
- Proper licensing for business operations

The system includes basic security measures to protect user data, such as authentication and validation. These measures help in maintaining data privacy and security.

Since the project does not involve real financial transactions or sensitive data, there are no major legal concerns. However, future enhancements such as payment gateway integration would require compliance with financial regulations.

Therefore, the system is legally feasible for academic use and can be made compliant with legal requirements for real-world implementation.

🔷 5. SCHEDULE FEASIBILITY

Schedule feasibility evaluates whether the project can be completed within the given time frame.

The Online Grocery Store project is designed to be completed within a limited time period, typically 2–3 months. The development process is divided into different phases, including planning, design, development, and testing.

The use of simple and well-known technologies ensures that the development process is smooth and efficient. The modular approach allows different parts of the system to be developed simultaneously, reducing the overall development time.

The project timeline is realistic and achievable. Each phase is assigned a specific duration, and proper planning ensures that the project is completed on time. Regular testing and debugging help in identifying and fixing issues early, preventing delays.

The availability of online resources and documentation also helps in speeding up the development process. Developers can easily find solutions to problems and implement features efficiently.

Therefore, the Online Grocery Store system is schedule feasible as it can be completed within the given time frame without any major delays.

🔷 6. SOCIAL FEASIBILITY

Social feasibility evaluates the impact of the system on society and how well it is accepted by users.

The Online Grocery Store system has a positive impact on society as it provides a convenient and efficient way of shopping. It reduces the need for physical travel, which can help in saving time and reducing traffic congestion.

The system is especially beneficial for elderly people, disabled individuals, and those with busy schedules. It provides them with an easy way to access essential goods without leaving their homes.

The system also supports digital transformation and promotes the use of technology in everyday life. It encourages people to adopt online platforms and improves digital literacy.

Overall, the system is socially feasible as it provides benefits to users and has a positive impact on society.

🏁 CONCLUSION OF FEASIBILITY STUDY

The feasibility study of the Online Grocery Store (ApnaMart) project clearly indicates that the system is practical, beneficial, and capable of being implemented successfully. The project is technically feasible as it uses available technologies, economically feasible due to low cost and high benefits, operationally feasible as it is user-friendly and efficient, legally feasible for academic purposes, and schedule feasible within the given time frame.

The system also has a positive social impact and supports digital transformation. Therefore, it can be concluded that the Online Grocery Store system is a viable and effective solution for modern grocery shopping needs.

🔍 SYSTEM ANALYSIS (Online Grocery Store – ApnaMart)

System analysis is a critical phase in the software development life cycle (SDLC) that involves studying the current system, identifying its limitations, and proposing a new system that overcomes those limitations. It focuses on understanding user requirements, system functionality, data flow, and overall system behavior. In the case of the Online Grocery Store (ApnaMart), system analysis helps in identifying the problems of traditional grocery shopping and designing an efficient online solution.

🔷 1. INTRODUCTION TO SYSTEM ANALYSIS

System analysis involves examining the existing system, understanding its weaknesses, and defining the requirements of the proposed system. It ensures that the new system is designed to meet user needs effectively and efficiently.

For the Online Grocery Store system, system analysis includes:

- Studying the traditional grocery shopping process
- Identifying problems in the existing system
- Defining requirements for the new system
- Designing system functionalities

🔷 2. EXISTING SYSTEM

The existing system refers to the traditional method of grocery shopping, where customers visit physical stores to purchase items.

🔶 Features of Existing System:

- Manual selection of products
- Physical visit to stores
- Cash or card payment at counters
- Limited product availability

🔶 Limitations of Existing System:

1. **Time-Consuming Process:**
   Customers need to travel to stores, search for products, and wait in queues.
2. **Limited Accessibility:**
   Stores operate only during specific hours, limiting access.
3. **Physical Effort Required:**
   Customers must carry heavy grocery items.
4. **Crowded Environment:**
   Shopping during peak hours can be stressful.
5. **Lack of Product Information:**
   Customers may not get complete details about products.
6. **No Home Delivery:**
   Customers must transport items themselves.

🔷 3. PROPOSED SYSTEM

The proposed system is the Online Grocery Store (ApnaMart), which provides a digital platform for grocery shopping.

🔶 Features of Proposed System:

- Online product browsing
- Add to cart functionality
- Secure login system
- Online order placement
- Admin management system

🔶 Advantages of Proposed System:

1. **Convenience:**
   Users can shop from home anytime.
2. **Time-Saving:**
   Reduces time spent on traveling and waiting.
3. **Wide Product Range:**
   Users can access multiple categories easily.

4. **Home Delivery:**
   Products are delivered to the user's address.
5. **User-Friendly Interface:**
   Easy navigation and simple design.
6. **Efficient Management:**
   Admin can manage products and orders easily.

🔷 4. SYSTEM REQUIREMENTS

System requirements define what is needed to develop and run the system.

🔶 Functional Requirements:

1. User Registration
2. User Login
3. Product Browsing
4. Add to Cart
5. Checkout
6. Order Management
7. Admin Login
8. Product Management
9. User Management

🔶 Non-Functional Requirements:

1. **Performance:** Fast response time
2. **Security:** Secure login and data protection
3. **Usability:** Easy-to-use interface
4. **Scalability:** Can handle more users
5. **Reliability:** System should work without errors

🔷 5. SYSTEM ARCHITECTURE

The Online Grocery Store system follows a **three-tier architecture**:

🔶 1. Presentation Layer:

- User interface (HTML, CSS, JavaScript)

🔶 2. Application Layer:

- Business logic (PHP)

🔶 3. Database Layer:

- Data storage (MySQL)

This architecture ensures separation of concerns and improves maintainability.

🔷 6. DATA FLOW ANALYSIS

Data flow analysis describes how data moves through the system.

🔶 Main Data Flow:

1. User inputs data (login, order details)
2. System processes data
3. Data is stored in database
4. Output is displayed to user

🔷 7. PROCESS ANALYSIS

The system includes several processes:

🔶 1. User Registration Process:

- User enters details
- Data is stored in database

🔶 2. Login Process:

- User enters credentials
- System verifies data

🔶 3. Product Selection:

- User browses products
- Adds items to cart

🔶 4. Order Processing:

- User places order
- Data stored in database

🔷 8. ENTITY ANALYSIS

Entities represent objects in the system.

🔶 Main Entities:

- User
- Product
- Order
- Cart
- Admin

Each entity has attributes such as ID, name, price, etc.

🔷 9. INPUT DESIGN

Input design focuses on how data is entered into the system.

🔶 Input Forms:

- Registration Form
- Login Form
- Checkout Form

These forms are designed to be simple and user-friendly.

🔷 10. OUTPUT DESIGN

Output design focuses on how information is displayed.

🔶 Outputs:

- Product list
- Order confirmation
- Admin dashboard

Outputs are displayed clearly for better understanding.

🔷 11. DATABASE ANALYSIS

The system uses MySQL database.

🔶 Tables:

- Users
- Products
- Orders
- Cart
- Admin

Relationships are maintained between tables.

🔷 12. SYSTEM DESIGN CONSIDERATIONS

🔶 Security:

- Login authentication
- Data validation

🔶 Performance:

- Fast loading pages
- Efficient queries

🔶 Scalability:

- Easy to add new features

🔷 13. USER INTERFACE ANALYSIS

The UI is designed to be:

- Simple
- Attractive
- Easy to navigate

Users can easily access all features.

🔷 14. CONSTRAINTS

🔶 Technical Constraints:

- Limited resources

🔶 Time Constraints:

- Project deadline

🔶 Financial Constraints:

- Low budget

🔷 15. ASSUMPTIONS

- Users have internet access
- Users know basic computer usage

🔷 16. RISK ANALYSIS

🔶 Possible Risks:

- System errors
- Data loss
- Security issues

🔶 Solutions:

- Regular testing
- Backup system
- Security measures

⚙️ SYSTEM SPECIFICATIONS & DESIGN (Online Grocery Store – ApnaMart)

🔷 1. INTRODUCTION

System specifications and design is a crucial phase in the development of any software system. It defines the technical requirements, system architecture, components, and overall design structure of the application. This phase ensures that the system is developed in a structured and organized manner, meeting all functional and non-functional requirements.

For the Online Grocery Store (ApnaMart), system specifications and design focus on defining the hardware and software requirements, system architecture, database design, user interface design, and module structure. This ensures that the system is efficient, scalable, and easy to maintain.

🔷 2. SYSTEM SPECIFICATIONS

System specifications describe the technical requirements needed to develop and run the application successfully.

🔶 2.1 HARDWARE REQUIREMENTS

The Online Grocery Store system requires basic hardware components for development and execution.

- **Processor:** Intel Core i3 or higher
- **RAM:** Minimum 4 GB (Recommended 8 GB)
- **Hard Disk:** Minimum 500 GB
- **Input Devices:** Keyboard and Mouse
- **Output Devices:** Monitor
- **Internet Connection:** Required for testing and deployment

These hardware requirements are sufficient for running the application smoothly without performance issues.

🔶 2.2 SOFTWARE REQUIREMENTS

The system uses the following software tools and technologies:

- **Operating System:** Windows 10 / Windows 11
- **Frontend Technologies:** HTML, CSS, JavaScript
- **Backend Technology:** PHP
- **Database:** MySQL
- **Server:** XAMPP (Apache Server)
- **Browser:** Google Chrome / Microsoft Edge
- **Code Editor:** Visual Studio Code

These tools are widely used, reliable, and suitable for web-based application development.

🔶 2.3 TECHNOLOGY STACK

The technology stack used in this project includes:

🔷 Frontend:

- HTML for structure
- CSS for styling
- JavaScript for interactivity

🔷 Backend:

- PHP for server-side scripting

🔷 Database:

- MySQL for data storage

This combination ensures a lightweight, efficient, and scalable system.

🔷 3. SYSTEM DESIGN

System design defines how the system components interact with each other. It includes architecture design, database design, and interface design.

🔶 3.1 SYSTEM ARCHITECTURE

The Online Grocery Store follows a **three-tier architecture**, which includes:

1. Presentation Layer:

- User interface (HTML, CSS, JS)
- Handles user interaction

2. Application Layer:

- Business logic (PHP)
- Processes user requests

3. Database Layer:

- Data storage (MySQL)
- Stores user, product, and order data

This architecture improves scalability, maintainability, and performance.

🔶 3.2 MODULE DESIGN

The system is divided into two main modules:

🔷 1. USER MODULE

This module allows customers to interact with the system.

Features:

- User Registration
- Login Authentication
- Product Browsing
- Add to Cart
- Checkout
- Order History

🔹 2. ADMIN MODULE

This module provides control over the system.

Features:

- Admin Login
- Add/Edit/Delete Products
- Manage Orders
- Manage Users
- View Reports

🔶 3.3 DATABASE DESIGN

The database is a key component of the system. It stores all data required for the application.

🔹 Tables Used:

1. **Users Table**
2. **Products Table**
3. **Orders Table**
4. **Cart Table**
5. **Admin Table**
6. **Categories Table**
7. **Payments Table**
8. **Feedback Table**

🔹 Example Table Structure:

```
CREATE TABLE products (
  id INT AUTO_INCREMENT PRIMARY KEY,
  name VARCHAR(100),
  price DECIMAL(10,2),
  category VARCHAR(50),
```

```
  image VARCHAR(255)
);
```

🔷 **Relationships:**

- User → places → Order
- Order → contains → Products
- Product → belongs → Category

🔶 3.4 INPUT DESIGN

Input design focuses on how users enter data into the system.

🔷 **Input Forms:**

1. **Registration Form**

- Name
- Email
- Password

2. **Login Form**

- Email
- Password

3. **Checkout Form**

- Address
- Phone Number
- Payment Method

🔷 **Features:**

- Simple layout
- Required field validation

- Error messages

🔶 3.5 OUTPUT DESIGN

Output design focuses on how information is displayed to users.

🔷 Outputs:

- Product Listing
- Cart Summary
- Order Confirmation
- Admin Dashboard

🔷 Features:

- Clear display
- Organized layout
- Real-time updates

🔶 3.6 USER INTERFACE DESIGN

The UI is designed to be:

- Attractive
- Simple
- Responsive

🔷 Pages:

- Home Page
- Product Page
- Cart Page
- Checkout Page

- Admin Dashboard

🔷 **Design Features:**

- Navigation bar
- Product cards
- Buttons and icons
- Responsive layout

🔶 3.7 PROCESS DESIGN

Process design defines how system processes work.

🔷 **Main Processes:**

1. User Registration
2. User Login
3. Product Selection
4. Cart Management
5. Order Processing

🔷 **Example Flow:**

User → Select Product → Add to Cart → Checkout → Order Stored

---

🔶 3.8 SECURITY DESIGN

Security is an important part of system design.

🔷 **Security Features:**

- User Authentication
- Password Protection

- Input Validation
- Session Management

🔶 3.9 PERFORMANCE DESIGN

Performance ensures the system runs efficiently.

🔷 **Techniques:**

- Optimized queries
- Fast loading pages
- Efficient data handling

🔶 3.10 SCALABILITY DESIGN

The system is designed to be scalable.

🔷 **Features:**

- Modular design
- Easy to add new features
- Supports more users

🔶 3.11 ERROR HANDLING

Error handling ensures system reliability.

🔷 **Features:**

- Input validation errors
- Database error handling
- User-friendly messages

🔶 3.12 NAVIGATION DESIGN

Navigation helps users move easily.

🔷 Features:

- Menu bar
- Search option
- Category filters

🔶 3.13 DEPLOYMENT DESIGN

Deployment defines how the system is launched.

🔷 Steps:

1. Install XAMPP
2. Place project in htdocs
3. Import database
4. Run project on localhost

🔶 3.14 SYSTEM WORKFLOW

🔷 Steps:

1. User registers/logins
2. Browses products
3. Adds items to cart
4. Places order
5. Admin manages order

🔶 3.15 DESIGN CONSTRAINTS

- Limited time

- Limited budget
- Basic technology

◆ 3.16 MAINTENANCE AND SUPPORT DESIGN

Maintenance is an important aspect of system design that ensures the system continues to function properly after deployment. The Online Grocery Store (ApnaMart) system is designed in a modular and flexible way so that it can be easily maintained and updated in the future.

The system supports different types of maintenance:

- Corrective Maintenance: Fixing bugs and errors identified after deployment.
- Adaptive Maintenance: Updating the system according to changes in technology or user requirements.
- Perfective Maintenance: Improving system performance and adding new features.

Regular maintenance ensures that the system remains efficient, secure, and up-to-date. Backup mechanisms can also be implemented to prevent data loss and ensure system reliability.

◆ 3.17 DATA MANAGEMENT DESIGN

Data management is a critical component of the system design. The Online Grocery Store system uses a structured database to store and manage all data efficiently. Data is organized into tables such as users, products, orders, and categories.

Key features of data management include:

- Data Integrity: Ensuring accurate and consistent data
- Data Security: Protecting user information
- Data Backup: Preventing data loss

- Efficient Retrieval: Fast access to data using queries

The system uses relational database concepts to maintain relationships between different tables, ensuring smooth data flow and consistency.

◆ 3.18 USER EXPERIENCE (UX) DESIGN

User experience plays a vital role in the success of any application. The ApnaMart system is designed to provide a smooth and enjoyable experience for users.

Key UX features include:

- Easy navigation between pages
- Fast loading speed
- Clear product display
- Simple checkout process

The system minimizes user effort and ensures that users can complete tasks quickly and efficiently. A good user experience increases user satisfaction and encourages repeated usage.

◆ 3.19 FUTURE DESIGN CONSIDERATIONS

The system is designed with future enhancements in mind. Some possible future improvements include:

- Integration of online payment gateways (UPI, Debit/Credit Card)
- Mobile application development
- Real-time order tracking system
- AI-based product recommendations
- Multi-language support

These enhancements can improve system functionality and make it more competitive in the market.

◆ 3.20 RELIABILITY AND AVAILABILITY DESIGN

The system is designed to be reliable and available to users at all times. It ensures that users can access the platform without interruptions.

Key features include:

- Stable server environment
- Error handling mechanisms
- Continuous availability (24/7 access)

This ensures that users can shop anytime without facing issues.

🏁 CONCLUSION

The system specifications and design of the Online Grocery Store (ApnaMart) project provide a clear structure for developing the application. The system is designed to be efficient, user-friendly, and scalable. It ensures smooth operation and meets all functional and technical requirements.

📅 PROJECT PLANNING & SCHEDULING (Online Grocery Store – ApnaMart)

🔷 1. INTRODUCTION

Project planning and scheduling is one of the most important phases in software development. It involves organizing tasks, allocating resources, defining timelines, and ensuring that the project is completed within the given time frame. Proper planning helps in avoiding delays, reducing risks, and ensuring the successful completion of the project.

In the Online Grocery Store (ApnaMart) project, planning and scheduling play a crucial role in managing different development phases such as requirement analysis, design, coding, testing, and deployment. This section explains how the project is planned, how tasks are divided, and how time is managed efficiently.

🔷 2. PROJECT PLANNING OVERVIEW

Project planning involves defining the scope, objectives, resources, and activities required to complete the project. It ensures that all tasks are clearly identified and executed in a systematic manner.

🔶 Key Planning Activities:

- Requirement gathering
- Resource allocation
- Task identification
- Time estimation
- Risk management

The main goal of project planning is to ensure that the project is completed successfully within the specified time and budget.

🔷 3. PROJECT DEVELOPMENT APPROACH

The Online Grocery Store system follows the **Software Development Life Cycle (SDLC)** approach. The SDLC provides a structured process for developing software.

◆ Phases of SDLC:

1. Requirement Analysis
2. System Design
3. Development
4. Testing
5. Deployment
6. Maintenance

Each phase is carefully planned and executed to ensure quality and efficiency.

◆ 4. PROJECT SCOPE

The scope of the project defines what is included and what is excluded.

◆ Included:

- User registration and login
- Product browsing
- Cart management
- Order placement
- Admin panel

◆ Excluded:

- Real payment gateway
- Advanced AI features

◆ 5. RESOURCE PLANNING

Resource planning involves identifying the resources required for the project.

◆ Human Resources:

- Developer (Student)
- Guide/Instructor

◆ Hardware Resources:

- Computer/Laptop
- Internet connection

◆ Software Resources:

- XAMPP
- VS Code
- Browser

◆ 6. TASK IDENTIFICATION

The project is divided into smaller tasks for easy management.

◆ Major Tasks:

1. Requirement Analysis
2. UI Design
3. Database Design
4. Coding (Frontend)
5. Coding (Backend)
6. Testing
7. Documentation

Each task is assigned a specific time duration.

◆ 7. TIME ESTIMATION

Time estimation helps in planning the project timeline.

◆ Estimated Duration:

Phase	Duration
Planning	1 week
Design	2 weeks
Development	4 weeks
Testing	2 weeks
Documentation	1 week

Total Duration: **10 weeks**

🔷 8. PROJECT SCHEDULING

Project scheduling defines when each task will be performed.

🔶 Weekly Schedule:

- Week 1 → Requirement Analysis
- Week 2–3 → System Design
- Week 4–7 → Development
- Week 8–9 → Testing
- Week 10 → Documentation

This schedule ensures systematic progress.

🔷 9. WORK BREAKDOWN STRUCTURE (WBS)

The Work Breakdown Structure divides the project into smaller parts.

🔶 WBS Structure:

1. Planning
    - Requirement Analysis
2. Design
    - UI Design
    - Database Design

3. Development
    - Frontend
    - Backend
4. Testing
5. Deployment
6. Documentation

This helps in better management and tracking.

🔷 10. GANTT CHART (Explanation)

A Gantt chart is used to represent project tasks over time.

🔶 **Example:**

- Planning → Week 1
- Design → Week 2–3
- Development → Week 4–7
- Testing → Week 8–9
- Documentation → Week 10

It helps in visualizing project progress.

🔷 11. PERT CHART (Explanation)

PERT (Program Evaluation and Review Technique) is used to analyze project tasks.

🔶 **Flow:**

Start → Planning → Design → Development → Testing → End

It helps in identifying dependencies between tasks.

🔷 12. RISK MANAGEMENT

Risk management involves identifying and handling risks.

🔶 **Possible Risks:**

1. Time Delay
2. Technical Issues
3. Data Loss
4. System Errors

🔶 **Risk Solutions:**

- Proper planning
- Regular testing
- Backup system
- Debugging

🔷 13. MONITORING AND CONTROL

Monitoring ensures that the project is progressing as planned.

🔶 **Activities:**

- Regular progress tracking
- Task completion checks
- Error correction

🔷 14. QUALITY ASSURANCE

Quality assurance ensures that the system meets required standards.

🔶 **Methods:**

- Testing
- Code review
- Error fixing

🔷 15. COMMUNICATION PLAN

Communication is important for project success.

🔶 Communication Methods:

- Regular meetings with guide
- Progress updates
- Feedback implementation

🔷 16. PROJECT DELIVERABLES

Deliverables are the outputs of the project.

🔶 Deliverables:

- Working system
- Project report
- Presentation (PPT)
- Source code

🔷 17. PROJECT CONSTRAINTS

🔶 Constraints:

- Limited time
- Limited budget
- Limited resources

🔷 18. ASSUMPTIONS

- Users have internet access
- System will run on standard devices

🔷 19. PROJECT WORKFLOW

🔶 **Steps:**

1. Requirement Analysis
2. Design
3. Development
4. Testing
5. Deployment

🔷 20. SUCCESS FACTORS

🔶 **Factors:**

- Proper planning
- Timely execution
- Good design
- Efficient coding

🔷 21. FUTURE PLANNING

Future planning includes system improvements.

🔶 **Enhancements:**

- Payment gateway
- Mobile app
- Delivery tracking

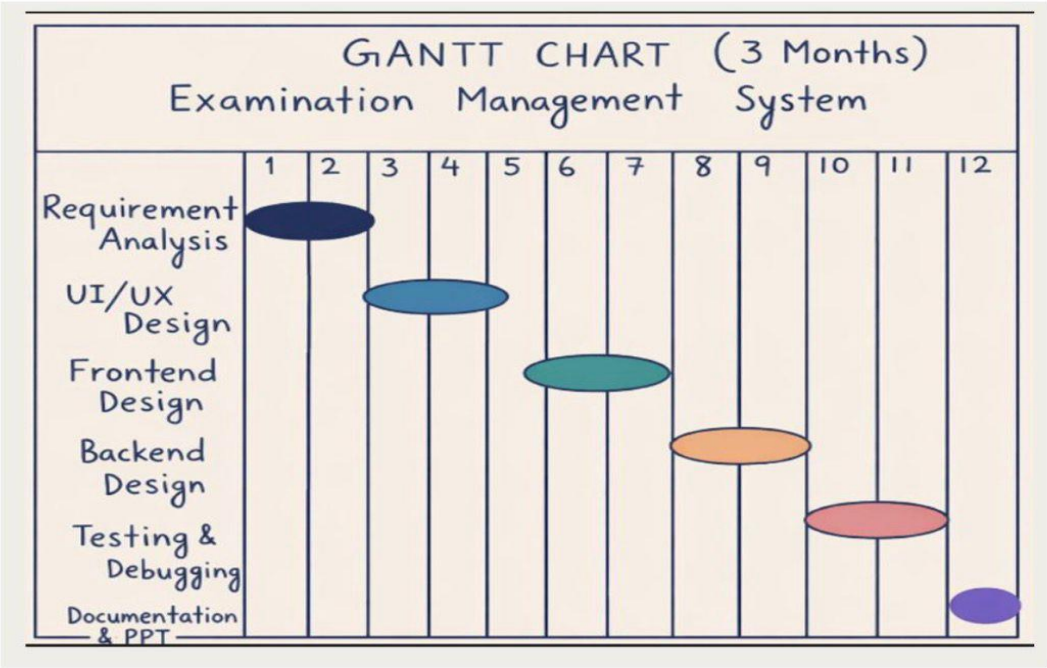
🏁 CONCLUSION

Project planning and scheduling for the Online Grocery Store (ApnaMart) ensure that the project is completed efficiently and within the given time frame. Proper planning, task division, and scheduling help in managing resources and reducing risks. The structured approach ensures the successful development of the system.

📊 GANTT CHART (Online Grocery Store – ApnaMart)

🔷 1. INTRODUCTION TO GANTT CHART

A Gantt Chart is a visual representation of a project schedule that shows the start and end dates of various tasks involved in a project. It is one of the most commonly used tools in project management for planning, scheduling, and tracking progress. The chart represents tasks on the vertical axis and time intervals on the horizontal axis.



Project Phases and Schedule

Task	Start Date	End Date	Duration
Requirement Analysis	15/12/2025	28/12/2025	2 Weeks
UI/UX Design	29/12/2025	11/01/2026	2 Weeks
Frontend Design	12/01/2026	25/01/2026	2 Weeks
Backend Design	26/01/2026	08/02/2026	2 Weeks
Testing & Debugging	09/02/2026	22/02/2026	2 Weeks
Documentation & PPT	23/02/2026	15/03/2026	3 Weeks

🔄 PERT CHART (Online Grocery Store – ApnaMart)

🔷 1. INTRODUCTION TO PERT CHART

PERT (Program Evaluation and Review Technique) is a project management tool used to analyze and represent the tasks involved in completing a project. It is particularly useful for planning, scheduling, and controlling complex projects where multiple activities are interdependent.

The PERT chart is a network diagram that shows the sequence of activities, their dependencies, and the time required to complete each activity. It helps project managers identify the critical path, which is the longest path in the project and determines the minimum time required to complete the project.

In the Online Grocery Store (ApnaMart) project, the PERT chart is used to represent the sequence of activities such as requirement analysis, system design, development, testing, and deployment. It helps in understanding how different tasks are connected and ensures that the project is completed efficiently within the given time frame.

🔷 2. OBJECTIVES OF PERT CHART

The main objectives of using a PERT chart in the project are:

- To identify all activities required to complete the project
- To determine the sequence and dependencies of tasks
- To estimate the time required for each activity
- To identify the critical path
- To improve project planning and scheduling

🔷 3. COMPONENTS OF PERT CHART

A PERT chart consists of the following components:

🔶 1. Events (Nodes):

Events represent the start or end of an activity. They are usually represented by circles in the diagram.

◆ 2. Activities (Arrows):

Activities represent the tasks that need to be completed. They are shown by arrows connecting two events.

◆ 3. Time Estimates:

Each activity has an associated time estimate, which indicates how long it will take to complete the task.

◆ 4. Dependencies:

Dependencies define the relationship between different activities. Some tasks cannot start until others are completed.

◆ 5. Critical Path:

The critical path is the longest sequence of activities in the project. It determines the minimum time required to complete the project.

◆ 4. PERT CHART IN ONLINE GROCERY STORE PROJECT

The Online Grocery Store project is divided into several activities, each with a specific duration and dependency.

◆ List of Activities:

Activity	Description	Duration
A	Requirement Analysis	1 week
B	System Design	2 weeks
C	Database Design	1 week

Activity	Description	Duration
D	Frontend Development	2 weeks
E	Backend Development	2 weeks
F	Integration	1 week
G	Testing	2 weeks
H	Documentation	1 week
I	Deployment	1 week

🔷 5. SEQUENCE OF ACTIVITIES

The sequence of activities in the project is as follows:

Start → A → B → C → D → E → F → G → H → I → End

🔷 6. NETWORK DIAGRAM EXPLANATION

The PERT network diagram represents the flow of activities from start to end.

🔶 Flow Explanation:

- The project starts with requirement analysis (A)
- After completion of A, system design (B) begins
- Database design (C) follows system design
- Frontend (D) and backend (E) development are carried out
- Integration (F) combines all components
- Testing (G) ensures system functionality
- Documentation (H) prepares project report
- Deployment (I) completes the project

🔷 7. CRITICAL PATH ANALYSIS

The critical path is the longest path in the project network.

◆ Critical Path:

A → B → C → D → E → F → G → H → I

◆ Total Duration:

1 + 2 + 1 + 2 + 2 + 1 + 2 + 1 + 1 = **13 weeks**

The critical path determines the minimum time required to complete the project. Any delay in these activities will delay the entire project.

◆ 8. IMPORTANCE OF CRITICAL PATH

- Helps in identifying important tasks
- Ensures timely project completion
- Helps in prioritizing activities
- Reduces delays

◆ 9. TIME ESTIMATION IN PERT

PERT uses three types of time estimates:

1. **Optimistic Time (O):** Minimum time required
2. **Most Likely Time (M):** Expected time
3. **Pessimistic Time (P):** Maximum time required

◆ Formula:

Expected Time (TE) = (O + 4M + P) / 6

This helps in accurate time estimation.

◆ 10. ADVANTAGES OF PERT CHART

1. Helps in project planning
2. Identifies task dependencies
3. Improves time management
4. Helps in risk management

5. Identifies critical activities

🔷 11. LIMITATIONS OF PERT CHART

1. Complex for large projects
2. Requires accurate time estimation
3. Difficult to update

🔷 12. ROLE IN PROJECT MANAGEMENT

PERT chart plays a key role in managing the project effectively. It helps in planning, scheduling, and controlling project activities.

🔷 13. PRACTICAL APPLICATION IN APNAMART

In the ApnaMart project, the PERT chart helps in organizing tasks and ensuring smooth workflow. It provides a clear understanding of task dependencies and helps in completing the project on time.

🔷 14. RISK MANAGEMENT USING PERT

PERT helps in identifying risks by analyzing task dependencies.

🔶 **Risks:**

- Delay in development
- Technical issues

🔶 **Solutions:**

- Proper planning
- Regular monitoring

🔷 15. COMPARISON WITH GANTT CHART

Feature	Gantt Chart	PERT Chart
Type	Bar chart	Network diagram
Focus	Time schedule	Task dependency
Complexity	Simple	Complex

🔷 16. FUTURE IMPROVEMENTS

- Use software tools for better visualization
- Dynamic updates
- Integration with project management tools

🏁 CONCLUSION

The PERT chart is an essential tool for project planning and scheduling. It provides a clear representation of tasks, dependencies, and timelines. In the Online Grocery Store (ApnaMart) project, the PERT chart helps in organizing activities, identifying the critical path, and ensuring timely completion of the project.

DFD (Data Flow Diagram) – Online Grocery Store (ApnaMart)

A Data Flow Diagram (DFD) is a graphical representation used in system analysis and design to illustrate how data flows through a system. It shows the movement of data between different components such as input sources, processes, data stores, and output destinations. DFD helps in understanding how information is processed within a system and how different parts of the system interact with each other.

It is one of the most important tools used by system analysts to visualize the logical flow of data without focusing on technical implementation details. The main purpose of a DFD is to provide a clear and structured view of the system, making it easier to identify processes, data transformations, and data storage points.

A DFD consists of four main elements:

- **External Entities: These are users or external systems that interact with the system, such as customers, admin, and payment gateway.**
- **Processes: These represent operations performed on data, such as user registration and login, product browsing, adding items to cart, order placement, and payment processing.**
- **Data Flows: These indicate the movement of data between different components of the system, such as customer details, product information, cart data, and order details.**
- **Data Stores: These represent places where data is stored, such as user database, product database, cart database, and order database.**

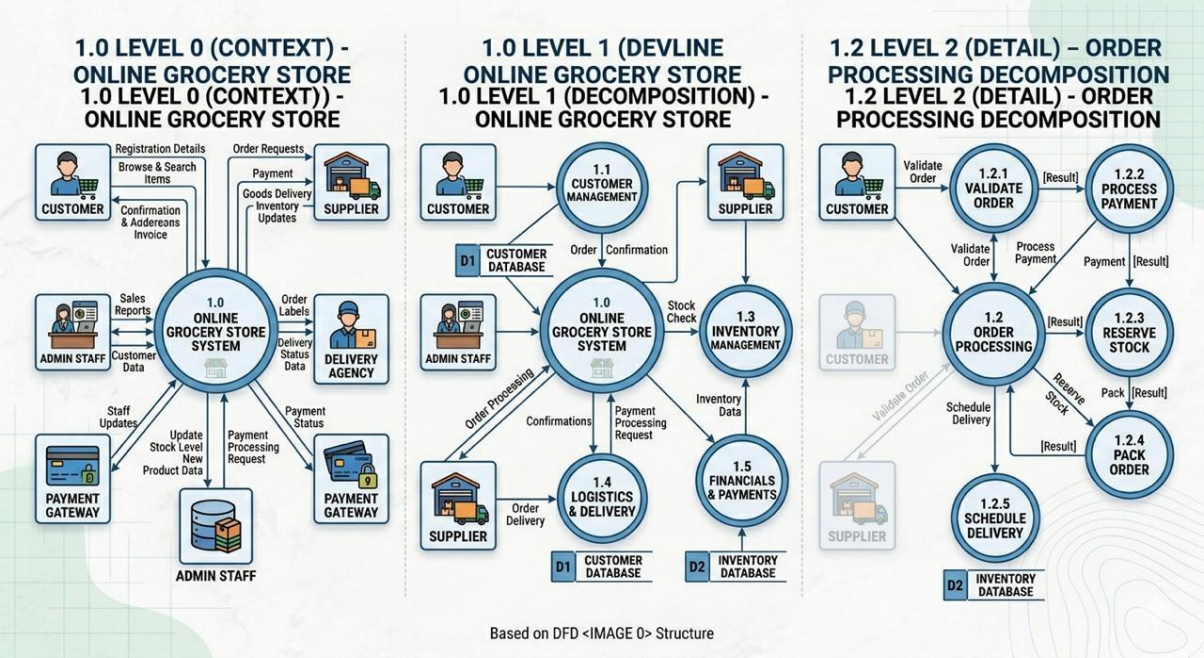
DFDs are designed in multiple levels such as Level 0, Level 1, and Level 2 to provide different levels of detail. The higher level (Level 0) shows an overall view of the system, while lower levels (Level 1 and Level 2) provide a more detailed breakdown of processes and data flow. This hierarchical approach helps in better understanding, analysis, and communication of system functionality.

In the context of the Online Grocery Store (ApnaMart), the DFD represents how customers interact with the system, how products are displayed, how items are added to the cart, how orders are placed, how payments are processed, and how all data is stored in the database. It ensures clarity, improves system design, and helps in identifying any errors or inefficiencies in the system.

DFD – Level 0:

DFD – Level 1:

DFD – Level 2:



🧩 MODULE DESCRIPTION (Online Grocery Store – ApnaMart)

🔷 1. INTRODUCTION

Module description is an important part of system design that explains how a project is divided into smaller functional units called modules. Each module performs a specific task and together all modules form a complete system.

Dividing a system into modules helps in better understanding, development, testing, and maintenance of the project.

In the Online Grocery Store (ApnaMart) project, the system is divided into multiple modules such as User Module, Admin Module, Product Module, Cart Module, Order Module, and Payment Module. Each module has its own functionality and responsibilities, which ensures smooth working of the system.

🔷 2. OVERVIEW OF MODULES

The Online Grocery Store system is mainly divided into the following modules:

1. User Module
2. Admin Module
3. Product Module
4. Cart Module
5. Order Module
6. Payment Module
7. Authentication Module
8. Feedback Module

Each module is explained in detail below.

🔷 3. USER MODULE

The User Module is one of the most important modules of the system. It allows customers to interact with the application and perform various operations related to grocery shopping.

🔶 Features of User Module:

- User Registration
- User Login
- View Products
- Search Products
- Add to Cart

- Place Order
- View Order History

🔶 **Working:**

The user first registers by entering basic details such as name, email, and password. After registration, the user can log in using their credentials. Once logged in, the user can browse products, add items to the cart, and place orders.

🔶 **Importance:**

This module provides the main interface for customers and ensures a smooth shopping experience.

🔷 4. ADMIN MODULE

The Admin Module provides control over the entire system. It is used by the administrator to manage products, users, and orders.

🔶 **Features of Admin Module:**

- Admin Login
- Add Product
- Edit Product
- Delete Product
- View Orders
- Manage Users

🔶 **Working:**

The admin logs into the system using secure credentials. After login, the admin can manage all system data such as adding new products, updating existing products, and removing products that are out of stock. The admin can also view customer orders and manage user accounts.

🔶 **Importance:**

This module ensures that the system runs smoothly and all data is properly managed.

🔷 5. PRODUCT MODULE

The Product Module is responsible for managing all grocery items available in the system.

🔶 Features:

- Add Product
- Update Product
- Delete Product
- View Product Details

🔶 Product Details Include:

- Product Name
- Price
- Category
- Image
- Description

🔶 Working:

Products are added by the admin and stored in the database. Users can view these products on the website and select items for purchase.

🔶 Importance:

This module ensures proper organization and display of products.

🔷 6. CART MODULE

The Cart Module manages the items selected by the user before placing an order.

🔶 Features:

- Add to Cart
- Remove from Cart
- Update Quantity
- View Total Price

🔶 **Working:**

When a user selects a product, it is added to the cart. The user can modify the cart by adding or removing items. The system calculates the total price automatically.

🔶 **Importance:**

This module provides flexibility to users and improves the shopping experience.

🔷 7. ORDER MODULE

The Order Module handles the process of placing and managing orders.

🔶 **Features:**

- Place Order
- View Order Details
- Order Confirmation
- Order History

🔶 **Working:**

After finalizing the cart, the user proceeds to checkout and enters delivery details. The system processes the order and stores it in the database. The user receives an order confirmation.

🔶 **Importance:**

This module is essential for completing the purchase process.

🔷 8. PAYMENT MODULE

The Payment Module manages payment-related activities.

🔶 **Features:**

- Select Payment Method
- Process Payment
- Payment Confirmation

🔶 **Working:**

Users can choose a payment method such as Cash on Delivery or Online Payment. The system records payment details and confirms the transaction.

🔶 **Importance:**

This module ensures smooth and secure transactions.

🔷 9. AUTHENTICATION MODULE

The Authentication Module ensures that only authorized users can access the system.

🔶 **Features:**

- User Login
- Password Protection
- Session Management

🔶 **Working:**

The system verifies user credentials during login and maintains session data to keep users logged in securely.

🔶 **Importance:**

This module provides security and protects user data.

🔷 10. FEEDBACK MODULE

The Feedback Module allows users to share their experience.

🔶 **Features:**

- Submit Feedback
- View Feedback

🔶 **Working:**

Users can provide feedback about products or services. This helps in improving the system.

🔶 **Importance:**

This module helps in enhancing user satisfaction.

🔷 11. MODULE INTERACTION

All modules are interconnected and work together to provide complete functionality.

🔶 Interaction Flow:

- User logs in → browses products → adds to cart → places order
- Admin manages products and orders
- Data is stored in database

This interaction ensures smooth system operation.

🔷 12. ADVANTAGES OF MODULAR DESIGN

- Easy to develop
- Easy to maintain
- Improves system performance
- Reduces complexity
- Allows future enhancements

🔷 13. FUTURE ENHANCEMENTS IN MODULES

- Add AI recommendation module
- Add delivery tracking module
- Add notification module

🏳 CONCLUSION

The module desc

ription of the Online Grocery Store (ApnaMart) project explains how the system is divided into different functional units. Each module performs a specific task and contributes to the overall functionality of the system. The modular design ensures that the system is efficient, scalable, and easy to maintain

📚 DATA DICTIONARY (Online Grocery Store – ApnaMart)

🔷 1. INTRODUCTION

A Data Dictionary is a structured collection of information that describes the data used in a system. It provides details about data elements such as their names, types, sizes, and descriptions. It acts as a reference guide for developers, designers, and users to understand how data is stored and used within the system.

In the Online Grocery Store (ApnaMart) project, the Data Dictionary defines all the data elements used in various modules such as user management, product management, cart management, and order processing. It ensures consistency, accuracy, and clarity in data handling.

The Data Dictionary helps in:

- Understanding database structure
- Maintaining data consistency
- Improving system documentation
- Reducing confusion during development

🔷 2. PURPOSE OF DATA DICTIONARY

The main purpose of the Data Dictionary is to provide detailed information about each data element used in the system.

🔶 Objectives:

- To define all data elements clearly
- To maintain consistency in naming
- To provide data type and size details
- To ensure proper data usage
- To help developers and users understand the system

🔷 3. DATA DICTIONARY STRUCTURE

Each data element in the Data Dictionary contains the following details:

- **Field Name** – Name of the data field
- **Data Type** – Type of data (INT, VARCHAR, etc.)
- **Size** – Length of the field
- **Description** – Meaning of the field

🔷 4. DATA DICTIONARY FOR TABLES

The Online Grocery Store system includes several tables. Each table is explained below with its fields.

🔶 4.1 USERS TABLE

This table stores information about users.

Field Name	Data Type	Size	Description
user_id	INT	11	Unique ID of user
name	VARCHAR	100	Name of user
email	VARCHAR	100	User email
password	VARCHAR	100	User password
phone	VARCHAR	15	Contact number
address	TEXT	—	User address

🔶 4.2 ADMIN TABLE

This table stores admin login details.

Field Name	Data Type	Size	Description
admin_id	INT	11	Unique admin ID
username	VARCHAR	100	Admin username
password	VARCHAR	100	Admin password

◆ 4.3 PRODUCTS TABLE

This table stores all grocery products.

Field Name	Data Type	Size	Description
product_id	INT	11	Unique product ID
name	VARCHAR	100	Product name
category	VARCHAR	50	Product category
price	DECIMAL	10,2	Product price
image	VARCHAR	255	Product image
description	TEXT	—	Product details
stock	INT	11	Available quantity

◆ 4.4 CATEGORIES TABLE

This table stores product categories.

Field Name	Data Type	Size	Description
category_id	INT	11	Category ID
category_name	VARCHAR	100	Category name

◆ 4.5 CART TABLE

This table stores user cart items.

Field Name	Data Type	Size	Description
cart_id	INT	11	Cart ID
user_id	INT	11	User ID
product_id	INT	11	Product ID
quantity	INT	11	Quantity

Field Name	Data Type	Size	Description
total_price	DECIMAL	10,2	Total cost

◆ 4.6 ORDERS TABLE

This table stores order details.

Field Name	Data Type	Size	Description
order_id	INT	11	Order ID
user_id	INT	11	User ID
order_date	DATE	—	Date of order
total_amount	DECIMAL	10,2	Total amount
status	VARCHAR	50	Order status

◆ 4.7 ORDER DETAILS TABLE

This table stores detailed order items.

Field Name	Data Type	Size	Description
order_detail_id	INT	11	Unique ID
order_id	INT	11	Order ID
product_id	INT	11	Product ID
quantity	INT	11	Quantity
price	DECIMAL	10,2	Product price

◆ 4.8 PAYMENTS TABLE

This table stores payment details.

Field Name	Data Type	Size	Description
payment_id	INT	11	Payment ID

Field Name	Data Type	Size	Description
order_id	INT	11	Order ID
amount	DECIMAL	10,2	Payment amount
payment_method	VARCHAR	50	Payment type
payment_status	VARCHAR	50	Status

◆ 4.9 FEEDBACK TABLE

This table stores user feedback.

Field Name	Data Type	Size	Description
feedback_id	INT	11	Feedback ID
user_id	INT	11	User ID
message	TEXT	—	Feedback message
date	DATE	—	Feedback date

◆ 5. DATA RELATIONSHIPS

Relationships between tables ensure proper data flow.

◆ Examples:

- User → Orders (One-to-Many)
- Order → Order Details (One-to-Many)
- Product → Cart (Many-to-One)
- Product → Category (Many-to-One)

These relationships help maintain data integrity.

◆ 6. DATA TYPES USED

Common data types used in the system:

- INT → Integer values

- VARCHAR → Text values
- DECIMAL → Price values
- DATE → Date values
- TEXT → Large text

🔷 7. DATA VALIDATION RULES

Data validation ensures accuracy.

🔶 **Rules:**

- Email must be valid format
- Password should not be empty
- Quantity must be positive
- Price must be numeric

🔷 8. DATA SECURITY

Data security is important to protect user information.

🔶 **Measures:**

- Password encryption
- Secure login system
- Input validation
- Session management

🔷 9. ADVANTAGES OF DATA DICTIONARY

- Improves data understanding
- Ensures consistency
- Helps in database design
- Useful for maintenance

🔷 10. LIMITATIONS

- Needs regular updates
- Can become complex for large systems

🏳 CONCLUSION

The Data Dictionary of the Online Grocery Store (ApnaMart) project provides detailed information about all data elements used in the system. It helps in understanding the database structure, maintaining consistency, and improving system documentation. A well-defined data dictionary ensures smooth development and efficient data management.

🔗 ER DIAGRAM (Entity Relationship Diagram) – Online Grocery Store (ApnaMart)

---

1. INTRODUCTION

An **Entity Relationship Diagram (ER Diagram)** is a graphical representation of the entities, attributes, and relationships present in a system. It is one of the most important parts of database design because it shows how data is connected and how different tables interact with each other. In any software project, an ER diagram helps in understanding the structure of the database in a simple and visual way.

In the **Online Grocery Store (ApnaMart)** project, the ER diagram explains how users, products, categories, carts, orders, payments, admin, and feedback are related to each other. Since this project is based on an online shopping system, data management is very important. The ER diagram helps in designing a database that is organized, efficient, and easy to manage.

The main purpose of the ER diagram is to represent the logical structure of the system before creating actual database tables. It helps developers understand what entities are required, what attributes should be included, and how entities should be connected. For students, it is also a very important part of the project report because it improves the quality of documentation and makes the project easier to explain during viva or presentation.

In this project, the ER diagram is designed in such a way that it supports all major system functions such as user registration, product browsing, cart management, order placement, payment processing, and admin control. It ensures proper data flow and helps avoid redundancy and inconsistency in the database.

---

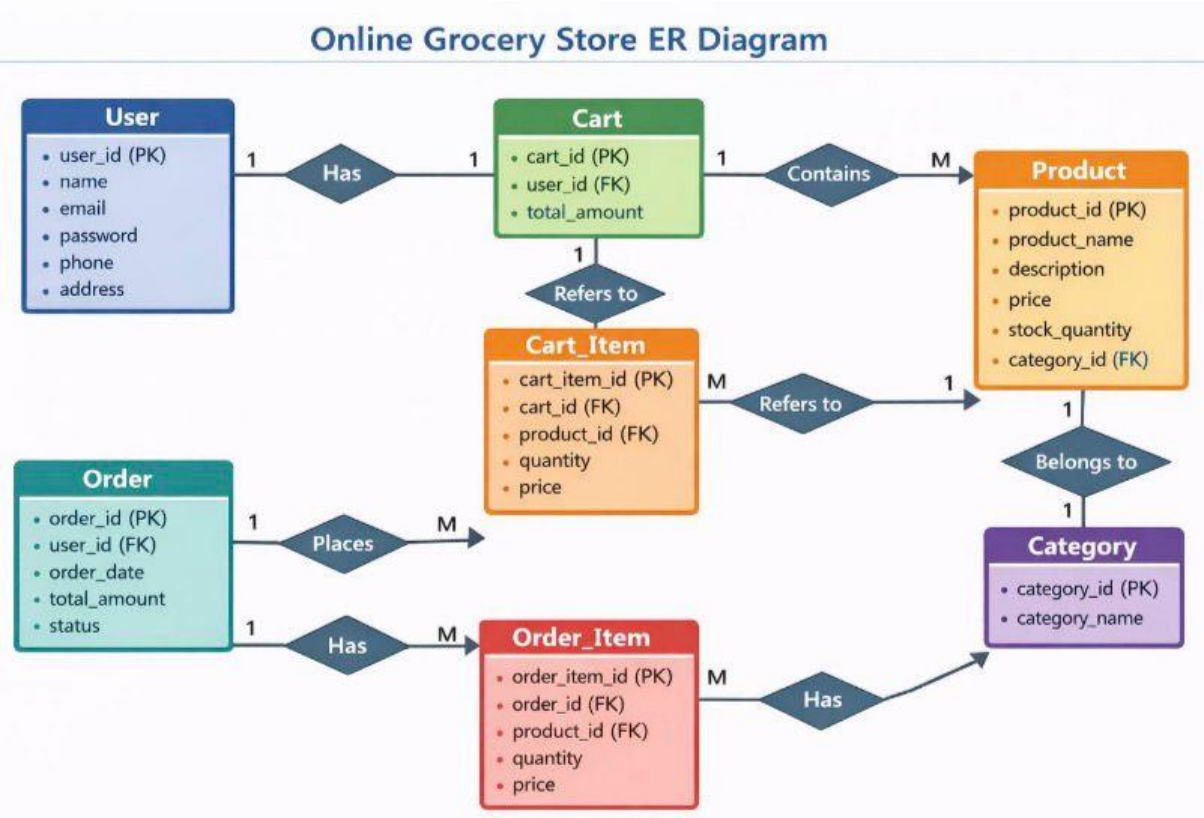
2. IMPORTANCE OF ER DIAGRAM

The ER diagram plays a very important role in system development and database design. It is useful in the following ways:

- It provides a clear understanding of the system database.
- It helps in identifying the required entities.
- It shows the relationships between different data tables.
- It reduces duplication of data.

- It improves database design and normalization.
- It helps developers during implementation.
- It makes the report more professional and easier to understand.

In the Online Grocery Store project, the ER diagram is important because the system involves many connected activities. For example, one user can place many orders, one order can contain multiple products, and one product can belong to one category. Without an ER diagram, these relationships may become confusing. Therefore, the ER diagram provides a structured and organized representation of the system.



🚀 SCOPE AND FUTURE ENHANCEMENT (Online Grocery Store – ApnaMart)

🔷 1. INTRODUCTION

The scope and future enhancement section describes the potential growth, usability, and possible improvements of the system. It explains how the current system can be used in real-world scenarios and what additional features can be added in the future to make it more advanced and efficient.

The Online Grocery Store (ApnaMart) system is designed as a basic e-commerce platform for purchasing grocery items online. While the current system provides essential functionalities such as user registration, product browsing, cart management, and order placement, there is a wide scope for further development and enhancement.

This section highlights the present capabilities of the system and explores various ways in which the system can be improved in the future.

🔷 2. SCOPE OF THE SYSTEM

The scope of the Online Grocery Store system defines the range of functionalities and applications that the system can handle. It describes how the system can be used in real-world environments and what benefits it offers to users and businesses.

🔶 2.1 Online Shopping Platform

The primary scope of the system is to provide an online platform for grocery shopping. Users can access the system through a web browser and purchase products without visiting physical stores.

🔶 **Benefits:**

- Saves time and effort
- Provides convenience
- Accessible anytime and anywhere

◆ 2.2 User Management System

The system allows users to register, log in, and manage their accounts.

◆ **Features:**

- User registration
- Login authentication
- Profile management

◆ **Scope:**

This feature can be used in any e-commerce platform for managing customers.

◆ 2.3 Product Management

The system provides functionality to manage grocery products.

◆ **Features:**

- Add products
- Update product details
- Delete products

◆ **Scope:**

This can be extended to manage inventory in real-time for businesses.

◆ 2.4 Cart and Order Management

The system allows users to add products to the cart and place orders.

◆ **Features:**

- Add to cart
- Update cart
- Place order

◆ **Scope:**

This functionality can be used in any online shopping application.

◆ 2.5 Admin Control System

The admin module allows administrators to control and manage the system.

◆ **Features:**

- Manage products
- View orders
- Manage users

◆ **Scope:**

This module can be used for business management and monitoring.

◆ 2.6 Database Management

The system uses a structured database to store data.

◆ **Scope:**

- Data storage
- Data retrieval
- Data security

This ensures smooth functioning of the system.

◆ 2.7 Educational Scope

The project has significant educational value.

◆ **Scope:**

- Helps students understand web development
- Demonstrates real-world application
- Improves technical skills

◆ 2.8 Business Scope

The system can be used by small and medium grocery businesses.

◆ Benefits:

- Increase customer reach
- Improve sales
- Reduce operational cost

◆ 3. LIMITATIONS OF CURRENT SYSTEM

Although the system provides basic functionality, it has some limitations:

- No real payment gateway integration
- No live order tracking
- Limited user interface design
- No mobile application
- Basic security features

These limitations provide opportunities for future enhancements.

◆ 4. FUTURE ENHANCEMENTS

Future enhancements refer to additional features that can be added to improve the system.

◆ 4.1 Online Payment Integration

Currently, the system may support basic payment methods like Cash on Delivery.

◆ Future Improvement:

- Integration with UPI, Debit Card, Credit Card
- Secure payment gateway

◆ Benefit:

- Faster and secure transactions

◆ 4.2 Mobile Application Development

The system is currently web-based.

◆ **Future Improvement:**

- Android and iOS mobile app

◆ **Benefit:**

- Better accessibility
- Improved user experience

◆ 4.3 Real-Time Order Tracking

Currently, users cannot track their orders.

◆ **Future Improvement:**

- Live tracking system

◆ **Benefit:**

- Improved customer satisfaction

◆ 4.4 AI-Based Recommendation System

The system currently does not provide recommendations.

◆ **Future Improvement:**

- Suggest products based on user behavior

◆ **Benefit:**

- Personalized shopping experience

◆ 4.5 Advanced Search and Filter

The current search functionality is basic.

◆ **Future Improvement:**

- Filter by price, category, rating

◆ **Benefit:**

- Better user experience

◆ 4.6 Multi-Language Support

The system currently supports only one language.

◆ **Future Improvement:**

- Support multiple languages

◆ **Benefit:**

- Wider audience reach

◆ 4.7 Notification System

The system does not provide notifications.

◆ **Future Improvement:**

- Email and SMS notifications

◆ **Benefit:**

- Keeps users updated

◆ 4.8 Delivery Management System

Currently, delivery is not managed in detail.

◆ **Future Improvement:**

- Delivery scheduling
- Delivery staff management

◆ **Benefit:**

- Better logistics management

◆ 4.9 Review and Rating System

Users cannot rate products.

◆ **Future Improvement:**

- Product reviews and ratings

◆ **Benefit:**

- Helps users make better decisions

◆ 4.10 Security Enhancements

Current security is basic.

◆ **Future Improvement:**

- Data encryption
- Secure authentication
- HTTPS implementation

◆ **Benefit:**

- Protects user data

◆ 4.11 Cloud Deployment

Currently, the system runs locally.

◆ **Future Improvement:**

- Deploy on cloud platforms

◆ **Benefit:**

- Scalability and reliability

◆ 4.12 Inventory Management System

The current system has basic stock management.

◆ **Future Improvement:**

- Real-time inventory tracking

🔶 **Benefit:**

- Avoids stock issues

🔷 5. FUTURE SCOPE IN INDUSTRY

The Online Grocery Store system has strong future potential in the industry.

🔶 **Trends:**

- Growth of e-commerce
- Increase in online shopping
- Demand for convenience

🔶 **Opportunities:**

- Startup business
- Expansion to large-scale platform
- Integration with delivery services

🔷 6. ADVANTAGES OF FUTURE ENHANCEMENTS

- Improves system functionality
- Enhances user experience
- Increases business value
- Provides competitive advantage

🏁 CONCLUSION

The scope of the Online Grocery Store (ApnaMart) project is very wide, as it can be used as a real-world e-commerce application. While the current system provides basic functionalities, there are many opportunities for future enhancements. By adding advanced features such as payment integration, mobile apps, AI recommendations, and delivery tracking, the system can be transformed into a complete and professional online grocery platform.

The project not only serves as an academic requirement but also has the potential to be developed into a real business solution.

🛠 IMPLEMENTATION METHODOLOGY (Online Grocery Store – ApnaMart)

🔷 1. INTRODUCTION

Implementation methodology refers to the step-by-step process used to develop and deploy a software system. It includes all stages from requirement analysis to final deployment and maintenance. A well-defined methodology ensures that the system is developed in a structured and efficient manner.

In the Online Grocery Store (ApnaMart) project, the implementation methodology follows a systematic approach based on the **Software Development Life Cycle (SDLC)**. This ensures that all phases of development are completed properly and the system meets user requirements.

🔷 2. SOFTWARE DEVELOPMENT LIFE CYCLE (SDLC)

The Online Grocery Store system is developed using the SDLC approach, which consists of the following phases:

1. Requirement Analysis
2. System Design
3. Development (Coding)
4. Testing
5. Deployment
6. Maintenance

Each phase plays an important role in the successful implementation of the system.

🔷 3. REQUIREMENT ANALYSIS

Requirement analysis is the first step in the implementation process. In this phase, the needs and expectations of users are identified.

🔶 Activities:

- Understanding user requirements

- Identifying system features
- Analyzing existing system
- Defining project scope

🔶 **Requirements Identified:**

- User registration and login
- Product browsing and search
- Cart management
- Order placement
- Admin panel

🔶 **Outcome:**

A clear understanding of what the system should do.

🔷 4. SYSTEM DESIGN

In this phase, the structure of the system is planned.

🔶 **Activities:**

- Designing system architecture
- Creating database design
- Designing user interface
- Preparing DFD and ER diagrams

🔶 **Design Components:**

- **Frontend Design:** HTML, CSS, JavaScript
- **Backend Design:** PHP
- **Database Design:** MySQL

🔶 **Outcome:**

A blueprint of the system that guides development.

🔷 5. DEVELOPMENT (CODING)

This is the main phase where the actual system is built.

🔶 **Frontend Development:**

- Designing web pages
- Creating user interface
- Adding interactivity using JavaScript

🔶 **Backend Development:**

- Handling user requests
- Processing data
- Connecting with database

🔶 **Database Implementation:**

- Creating tables
- Writing SQL queries
- Managing data

🔶 **Tools Used:**

- Visual Studio Code
- XAMPP Server
- MySQL

🔶 **Outcome:**

A working system with all functionalities implemented.

🔷 6. TESTING

Testing ensures that the system works correctly and is free from errors.

🔶 **Types of Testing:**

1. **Unit Testing:** Testing individual modules
2. **Integration Testing:** Testing combined modules
3. **System Testing:** Testing the complete system

🔶 **Testing Activities:**

- Checking login functionality
- Verifying cart operations
- Testing order placement
- Debugging errors

🔶 **Outcome:**

A stable and error-free system.

🔷 7. DEPLOYMENT

Deployment is the process of making the system available for use.

🔶 **Steps:**

1. Install XAMPP
2. Place project files in htdocs
3. Import database
4. Run system on browser

🔶 **Outcome:**

The system becomes accessible to users.

🔷 8. MAINTENANCE

Maintenance ensures that the system continues to work properly after deployment.

🔶 **Types:**

- **Corrective Maintenance:** Fixing errors
- **Adaptive Maintenance:** Updating system
- **Perfective Maintenance:** Adding features

🔶 **Outcome:**

Improved system performance and reliability.

🔷 9. IMPLEMENTATION TOOLS AND TECHNOLOGIES

The project uses the following tools:

🔶 **Frontend:**

- HTML
- CSS
- JavaScript

🔶 **Backend:**

- PHP

🔶 **Database:**

- MySQL

🔶 **Server:**

- XAMPP

🔶 **Editor:**

- VS Code

These technologies are simple, efficient, and suitable for web development.

🔷 10. IMPLEMENTATION STRATEGY

The system is developed using a modular approach.

🔶 **Steps:**

1. Develop user module
2. Develop admin module
3. Develop product module
4. Integrate all modules
5. Test the system

🔶 **Advantage:**

- Easy development

- Easy debugging
- Better performance

🔷 11. DATA IMPLEMENTATION

Data implementation involves storing and managing data.

🔶 **Activities:**

- Creating database tables
- Defining relationships
- Writing SQL queries

🔶 **Example Tables:**

- Users
- Products
- Orders
- Cart

🔶 **Outcome:**

Efficient data management system.

🔷 12. USER INTERFACE IMPLEMENTATION

The UI is designed to be simple and user-friendly.

🔶 **Features:**

- Navigation bar
- Product display
- Buttons and forms

🔶 **Outcome:**

Better user experience.

🔷 13. SECURITY IMPLEMENTATION

Security is implemented to protect user data.

🔶 **Measures:**

- Login authentication
- Input validation
- Session management

🔶 **Outcome:**

Safe and secure system.

🔷 14. CHALLENGES FACED

During implementation, some challenges may arise:

- Database connection issues
- Coding errors
- UI design problems

🔶 **Solutions:**

- Debugging
- Testing
- Proper planning

🔷 15. ADVANTAGES OF IMPLEMENTATION METHODOLOGY

- Structured development process
- Easy to manage
- Reduces errors
- Improves quality

🔷 16. FUTURE IMPLEMENTATION IMPROVEMENTS

- Add payment gateway
- Improve UI design

- Develop mobile app

🏳 CONCLUSION

The implementation methodology of the Online Grocery Store (ApnaMart) project follows a structured approach based on SDLC. Each phase, from requirement analysis to maintenance, plays an important role in the development of the system. The use of proper tools, technologies, and strategies ensures that the system is efficient, user-friendly, and reliable.

🧪 TESTING (Online Grocery Store – ApnaMart)

🔷 1. INTRODUCTION

Testing is a crucial phase in the software development life cycle (SDLC). It is the process of evaluating a system or application to ensure that it works correctly and meets the specified requirements. The main goal of testing is to identify and fix errors, bugs, or defects before the system is deployed for use.

In the Online Grocery Store (ApnaMart) project, testing plays a vital role in ensuring that all functionalities such as user registration, login, product browsing, cart management, and order placement work smoothly. Testing ensures that the system is reliable, efficient, and user-friendly.

🔷 2. OBJECTIVES OF TESTING

The main objectives of testing in this project are:

- To ensure that the system works as expected
- To identify and fix errors or bugs
- To verify all functionalities
- To improve system performance
- To ensure data accuracy and security

🔷 3. TYPES OF TESTING

Different types of testing are performed in the Online Grocery Store system to ensure complete functionality.

🔶 3.1 UNIT TESTING

Unit testing involves testing individual modules or components of the system.

🔷 **Example:**

- Testing login functionality
- Testing add to cart feature

🔹 **Purpose:**

To ensure that each module works correctly.

🔸 3.2 INTEGRATION TESTING

Integration testing checks whether different modules work together properly.

🔹 **Example:**

- Login + Cart + Order processing

🔹 **Purpose:**

To verify interaction between modules.

🔸 3.3 SYSTEM TESTING

System testing evaluates the complete system.

🔹 **Example:**

- Full shopping process (login → add to cart → order)

🔹 **Purpose:**

To ensure overall system functionality.

🔸 3.4 USER ACCEPTANCE TESTING (UAT)

This testing is performed by users to ensure the system meets their needs.

🔹 **Purpose:**

To confirm user satisfaction.

🔸 3.5 PERFORMANCE TESTING

Performance testing checks system speed and efficiency.

🔹 **Purpose:**

To ensure fast response time.

🔶 3.6 SECURITY TESTING

Security testing ensures data protection.

🔷 **Purpose:**

To prevent unauthorized access.

🔷 4. TEST CASES

Test cases are used to verify system functionality.

🔶 4.1 Login Test Case

Test Case	Input	Expected Output
Valid Login	Correct email/password	Login success
Invalid Login	Wrong password	Error message

🔶 4.2 Registration Test Case

Test Case	Input	Expected Output
Valid Registration	Proper details	Account created
Invalid Email	Wrong format	Error

🔶 4.3 Add to Cart Test Case

Test Case	Action	Expected Output
Add Product	Click add	Product added
Remove Product	Click remove	Product removed

🔶 4.4 Order Test Case

Test Case	Action	Expected Output
Place Order	Checkout	Order success
Empty Cart	Checkout	Error message

🔷 5. TESTING PROCESS

The testing process follows these steps:

1. Test planning
2. Test case design
3. Test execution
4. Result analysis
5. Bug fixing

🔷 6. BUGS AND ERROR HANDLING

During testing, some errors may occur.

🔶 Common Errors:

- Login failure
- Database connection error
- Cart calculation error

🔶 Solutions:

- Debugging
- Code correction
- Retesting

🔷 7. VALIDATION AND VERIFICATION

🔶 Validation:

Ensures the system meets user needs.

🔶 **Verification:**

Ensures the system is built correctly.

🔷 8. TESTING TOOLS

Tools used in testing:

- Browser (Chrome/Edge)
- XAMPP Server
- Manual testing

🔷 9. TESTING ENVIRONMENT

The system is tested in a local environment.

🔶 **Setup:**

- Windows OS
- XAMPP
- MySQL
- Browser

🔷 10. PERFORMANCE EVALUATION

The system is evaluated based on:

- Response time
- Load handling
- Efficiency

🔷 11. SECURITY CHECKS

Security measures tested:

- Login authentication
- Data validation

- Session handling

🔷 12. ADVANTAGES OF TESTING

- Improves system quality
- Reduces errors
- Ensures reliability
- Enhances user satisfaction

🔷 13. LIMITATIONS OF TESTING

- Time-consuming
- Cannot guarantee 100% error-free system
- Depends on test cases

🔷 14. FUTURE TESTING IMPROVEMENTS

- Automated testing
- Advanced security testing
- Load testing tools

🏳 CONCLUSION

Testing is an essential part of the Online Grocery Store (ApnaMart) project. It ensures that the system is reliable, efficient, and user-friendly. By performing different types of testing such as unit testing, integration testing, and system testing, the project achieves high quality and performance. Testing helps in identifying errors and improving the system before deployment.

⚠️ LIMITATIONS AND FURTHER ENHANCEMENTS (Online Grocery Store – ApnaMart)

🔷 1. INTRODUCTION

Every software system has certain limitations, especially when it is developed as an academic project with limited time, resources, and scope. Identifying these limitations is important because it helps in understanding the areas where the system can be improved. Along with limitations, further enhancements provide a roadmap for future development and improvement of the system.

The Online Grocery Store (ApnaMart) system is designed to provide basic functionalities such as user registration, product browsing, cart management, and order placement. While the system works effectively for its intended purpose, it still has some limitations that can be addressed in the future.

🔷 2. LIMITATIONS OF THE SYSTEM

The current version of the Online Grocery Store system has the following limitations:

🔶 2.1 Limited Payment Options

The system may only support basic payment methods such as Cash on Delivery.

Issue:

- No support for online payments like UPI, debit card, or credit card

Impact:

- Limits user convenience
- Reduces system usability

🔶 2.2 No Real-Time Order Tracking

The system does not provide live tracking of orders.

Issue:

- Users cannot see the status of delivery in real time

Impact:

- Reduced transparency
- Lower user satisfaction

◆ 2.3 Basic User Interface

The user interface of the system is simple and may lack advanced design features.

Issue:

- Limited animations and modern UI elements

Impact:

- Less attractive user experience

◆ 2.4 No Mobile Application

The system is only web-based.

Issue:

- No Android or iOS app

Impact:

- Limited accessibility

◆ 2.5 Limited Security Features

The system includes only basic security measures.

Issue:

- No advanced encryption
- No multi-factor authentication

Impact:

- Potential security risks

◆ 2.6 No Product Review System

Users cannot rate or review products.

Issue:

- No feedback mechanism for products

Impact:

- Users cannot make informed decisions

◆ 2.7 No Notification System

The system does not provide notifications.

Issue:

- No email or SMS updates

Impact:

- Users are not informed about order status

◆ 2.8 Limited Scalability

The system is designed for small-scale use.

Issue:

- Cannot handle a large number of users

Impact:

- Not suitable for large businesses

◆ 2.9 Manual Delivery Management

Delivery system is not automated.

Issue:

- No delivery tracking or scheduling

Impact:

- Inefficient logistics

🔶 2.10 Basic Search Functionality

Search functionality is limited.

Issue:

- No advanced filters

Impact:

- Users may find it difficult to search products

🔷 3. FURTHER ENHANCEMENTS

To overcome the above limitations, several enhancements can be implemented in the future.

🔶 3.1 Online Payment Integration

Future versions can include:

- UPI payment
- Credit/Debit card
- Digital wallets

Benefit:

- Faster and secure transactions
- Improved user experience

🔶 3.2 Real-Time Order Tracking

Add a tracking system that shows:

- Order status
- Delivery location

Benefit:

- Increased transparency

- Better customer satisfaction

◆ 3.3 Improved User Interface

Enhancements in UI can include:

- Modern design
- Animations
- Better layout

Benefit:

- Attractive and user-friendly system

◆ 3.4 Mobile Application Development

Develop mobile apps for:

- Android
- iOS

Benefit:

- Easy access
- Increased user engagement

◆ 3.5 Advanced Security Features

Security can be improved by:

- Data encryption
- Two-factor authentication
- Secure login system

Benefit:

- Better protection of user data

◆ 3.6 Product Review and Rating System

Add features for:

- Product ratings
- Customer reviews

Benefit:

- Helps users make better decisions

◆ 3.7 Notification System

Implement notifications such as:

- Order confirmation
- Delivery updates
- Offers and discounts

Benefit:

- Keeps users informed

◆ 3.8 Cloud Deployment

Deploy system on cloud platforms.

Benefit:

- Better scalability
- Improved performance

◆ 3.9 Inventory Management System

Enhance stock management:

- Real-time stock updates
- Low stock alerts

Benefit:

- Better inventory control

🔶 3.10 AI-Based Recommendation System

Add AI features:

- Suggest products based on user behavior

Benefit:

- Personalized experience

🔶 3.11 Multi-Language Support

Add support for multiple languages.

Benefit:

- Wider user reach

🔶 3.12 Delivery Management System

Improve delivery system:

- Delivery scheduling
- Delivery staff tracking

Benefit:

- Efficient logistics

🔷 4. BENEFITS OF FUTURE ENHANCEMENTS

- Improves system performance
- Enhances user experience
- Increases business value
- Provides competitive advantage

🔷 5. FUTURE SCOPE IN REAL-WORLD APPLICATION

The Online Grocery Store system has great potential in real-world applications.

🔶 Opportunities:

- Startup business
- Expansion into large-scale platform
- Integration with delivery services

🏁 CONCLUSION

The Online Grocery Store (ApnaMart) system, while functional and efficient, has certain limitations due to its basic design and academic scope. However, these limitations can be addressed through various future enhancements such as payment integration, mobile applications, real-time tracking, and advanced security features.

By implementing these improvements, the system can be transformed into a fully functional and professional e-commerce platform. The project not only fulfills academic requirements but also has the potential to be developed into a real-world business solution.

🔐 SECURITY MECHANISM (Online Grocery Store – ApnaMart)

🔷 1. INTRODUCTION

Security is one of the most important aspects of any web-based application. In an Online Grocery Store system like ApnaMart, users provide sensitive information such as personal details, login credentials, and sometimes payment-related data. Therefore, it is essential to implement proper security mechanisms to protect user data and ensure safe system operation.

The Security Mechanism refers to the set of techniques and methods used to protect the system from unauthorized access, data breaches, and cyber threats. A secure system not only protects data but also builds trust among users.

🔷 2. OBJECTIVES OF SECURITY MECHANISM

The main objectives of implementing security in the system are:

- To protect user data from unauthorized access
- To ensure secure login and authentication
- To prevent data tampering and hacking
- To maintain data confidentiality and integrity
- To provide a safe user experience

🔷 3. TYPES OF SECURITY MECHANISMS USED

The Online Grocery Store system uses several basic security mechanisms:

🔶 3.1 USER AUTHENTICATION

User authentication ensures that only authorized users can access the system.

🔷 **Implementation:**

- Login system using email and password
- Verification of user credentials from database

🔷 **Benefit:**

- Prevents unauthorized access

🔶 3.2 PASSWORD SECURITY

Passwords are sensitive information and must be protected.

🔷 **Implementation:**

- Passwords should be stored in encrypted form (e.g., hashing)
- Avoid storing plain text passwords

🔷 **Benefit:**

- Protects user accounts

🔶 3.3 SESSION MANAGEMENT

Session management helps maintain user login status.

🔷 **Implementation:**

- Use PHP sessions
- Store user_id in session
- Logout option to destroy session

🔷 **Benefit:**

- Secure user session handling

🔶 3.4 INPUT VALIDATION

Input validation ensures that the data entered by users is correct and safe.

🔷 **Implementation:**

- Validate email format
- Check required fields
- Restrict invalid characters

🔷 **Benefit:**

- Prevents malicious inputs

🔶 3.5 SQL INJECTION PREVENTION

SQL Injection is a common attack where hackers manipulate database queries.

🔷 **Implementation:**

- Use mysqli_real_escape_string()
- Use prepared statements

🔷 **Benefit:**

- Protects database from attacks

🔶 3.6 ACCESS CONTROL

Access control restricts system access based on user roles.

🔷 **Implementation:**

- Separate admin and user roles
- Admin panel access only for admin

🔷 **Benefit:**

- Prevents unauthorized operations

🔶 3.7 DATA CONFIDENTIALITY

Confidentiality ensures that sensitive data is not exposed.

🔷 **Implementation:**

- Protect user details
- Avoid displaying sensitive data

🔷 **Benefit:**

- Maintains privacy

🔶 3.8 ERROR HANDLING

Proper error handling prevents exposure of system details.

🔷 **Implementation:**

- Show user-friendly error messages
- Hide database errors

🔷 **Benefit:**

- Prevents attackers from gaining system information

🔷 4. ADDITIONAL SECURITY MEASURES (FUTURE)

To improve security further, the following features can be added:

🔶 4.1 HTTPS IMPLEMENTATION

- Secure communication using SSL
- Encrypt data transfer

🔶 4.2 TWO-FACTOR AUTHENTICATION

- OTP verification
- Extra layer of security

🔶 4.3 CAPTCHA SYSTEM

- Prevent bots
- Secure forms

🔶 4.4 DATA ENCRYPTION

- Encrypt sensitive data
- Use secure algorithms

🔶 4.5 FIREWALL PROTECTION

- Protect server from attacks

🔷 5. ADVANTAGES OF SECURITY MECHANISMS

- Protects user data
- Prevents unauthorized access
- Improves system reliability
- Builds user trust

🔷 6. LIMITATIONS

- Basic security in current system
- Needs improvement for large-scale use

🏳 CONCLUSION

Security mechanisms are essential for the Online Grocery Store (ApnaMart) system. By implementing authentication, password security, session management, and input validation, the system ensures a basic level of protection. However, advanced security features can be added in the future to make the system more robust and secure. A secure system not only protects data but also increases user confidence and reliability.

DATABASE MODEL (Online Grocery Store – ApnaMart)

1. Introduction

A database model is one of the most important parts of any software project because it defines how data is stored, organized, and managed within the system. In the Online Grocery Store (ApnaMart) project, the database model plays a major role in maintaining user details, product information, cart data, order records, payment details, and admin management.

The Online Grocery Store system is a web-based application, and for any e-commerce project, data management is very important. The database ensures that all information entered by users and administrators is stored safely and can be retrieved whenever needed. Without a proper database model, the system cannot function efficiently.

In this project, the database is designed using **MySQL**, which is a relational database management system. A relational database stores data in the form of tables, where each table contains rows and columns. Each table represents a specific entity, such as users, products, orders, or payments. Relationships are also created between tables so that the system can manage connected data smoothly.

The database model of ApnaMart is designed to support all the important features of the system, such as user registration, login, product display, add to cart, order placement, payment handling, and admin control. The database model ensures accuracy, consistency, and proper data flow across the system.

---

2. Purpose of Database Model

The main purpose of the database model in the Online Grocery Store project is to store and manage all information required for the system.

Main purposes are:

- To store user details securely
- To maintain product information
- To manage shopping cart data
- To record order details
- To handle payment information
- To help the admin manage the system

- To provide accurate and fast retrieval of data

A well-designed database model improves system performance, reduces data duplication, and supports future enhancements.

---

3. Type of Database Used

The Online Grocery Store project uses a **Relational Database Model**.

Why relational database is used:

- Easy to create tables
- Supports relationships between data
- Easy to query using SQL
- Provides accuracy and consistency
- Best for web-based applications

MySQL is used because it is simple, fast, secure, and widely used in PHP projects. It works smoothly with XAMPP and is suitable for student-level as well as real-world projects.

---

4. Main Tables in Database Model

The database model of the Online Grocery Store project contains several important tables. These tables store all required data for the system.

Main tables are:

1. Users
2. Admin
3. Categories
4. Products
5. Cart
6. Orders
7. Order Details
8. Payments
9. Feedback

Each table has a specific purpose and contributes to the overall working of the project.

---

5. Description of Tables

5.1 Users Table

This table stores all customer-related information.

Fields:

- user_id
- name
- email
- password
- phone
- address

Purpose:

It helps the system identify each customer and allows users to register and log in.

5.2 Admin Table

This table stores the admin login details.

Fields:

- admin_id
- username
- password

Purpose:

It allows the admin to log in securely and manage the system.

5.3 Categories Table

This table stores the categories of products.

Fields:

- category_id
- category_name

Purpose:

It helps organize products into groups such as fruits, vegetables, dairy, beverages, etc.

5.4 Products Table

This table stores all information related to grocery products.

Fields:

- product_id
- product_name
- price
- image
- description
- stock
- category_id

Purpose:

It is used to display products to users and maintain stock information.

5.5 Cart Table

This table stores products added to the shopping cart.

Fields:

- cart_id
- user_id
- product_id
- quantity
- total_price

Purpose:

It temporarily stores user-selected items before order placement.

5.6 Orders Table

This table stores order information.

Fields:

- order_id
- user_id
- order_date
- total_amount
- order_status
- address
- payment_method

Purpose:

It records all customer orders placed through the system.

5.7 Order Details Table

This table stores detailed information about products included in each order.

Fields:

- order_detail_id
- order_id
- product_id
- quantity
- price

Purpose:

It helps store multiple products under one order.

5.8 Payments Table

This table stores payment details.

Fields:

- payment_id
- order_id
- amount
- payment_method
- payment_status

Purpose:

It records payment transactions and status.

5.9 Feedback Table

This table stores customer feedback.

Fields:

- feedback_id
- user_id
- message
- date

Purpose:

It allows users to submit suggestions and comments.

---

6. Relationships Between Tables

The database model is relational, so relationships are created between tables.

Important relationships:

- One user can place many orders
- One user can have many cart items
- One category can contain many products

- One order can have many order details
- One order has one payment
- One user can give many feedbacks

These relationships help in maintaining proper connectivity between tables and ensure smooth data handling.

---

7. Primary Key and Foreign Key

Primary Key

A primary key is a unique field in a table that identifies each record.

Examples:

- user_id in Users table
- product_id in Products table
- order_id in Orders table

Foreign Key

A foreign key is a field that creates a relationship between two tables.

Examples:

- user_id in Orders table refers to Users table
- category_id in Products table refers to Categories table
- order_id in Payments table refers to Orders table

Primary keys and foreign keys are very important for maintaining data integrity.

---

8. Advantages of Database Model

The database model of the Online Grocery Store project provides many advantages:

- Organized storage of data
- Easy retrieval of information
- Reduced data duplication
- Better data integrity

- Fast processing of user requests
- Easy management by admin
- Supports future expansion

The relational model makes the system more efficient and reliable.

---

9. Limitations of Current Database Model

Although the database model works well for the academic project, it has some limitations:

- It is suitable mainly for small and medium-scale systems
- Advanced optimization is not included
- No complex indexing or backup mechanism
- Limited transaction security in current version

These limitations can be improved in future versions.

---

10. Conclusion

The database model of the Online Grocery Store (ApnaMart) project is the backbone of the entire system. It ensures that all important information such as users, products, cart items, orders, and payments is stored and managed properly. The use of MySQL as a relational database makes the system efficient, structured, and easy to maintain.

The tables and relationships are designed in such a way that the project can handle all major e-commerce functionalities smoothly. Overall, the database model provides a strong foundation for the Online Grocery Store system and plays a key role in its successful operation.

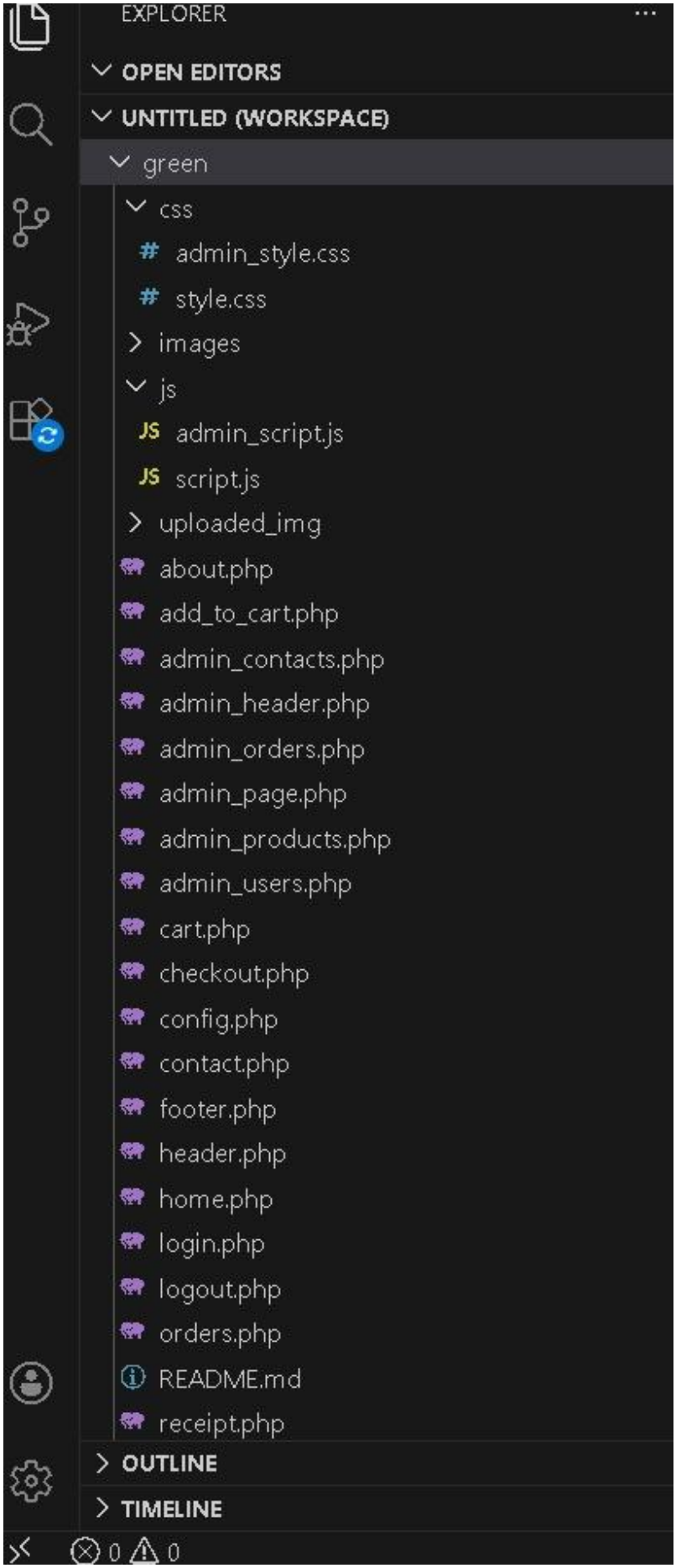
🖼 FILES OF THE PROJECT (Online Grocery Store – ApnaMart)

🔷 1. INTRODUCTION

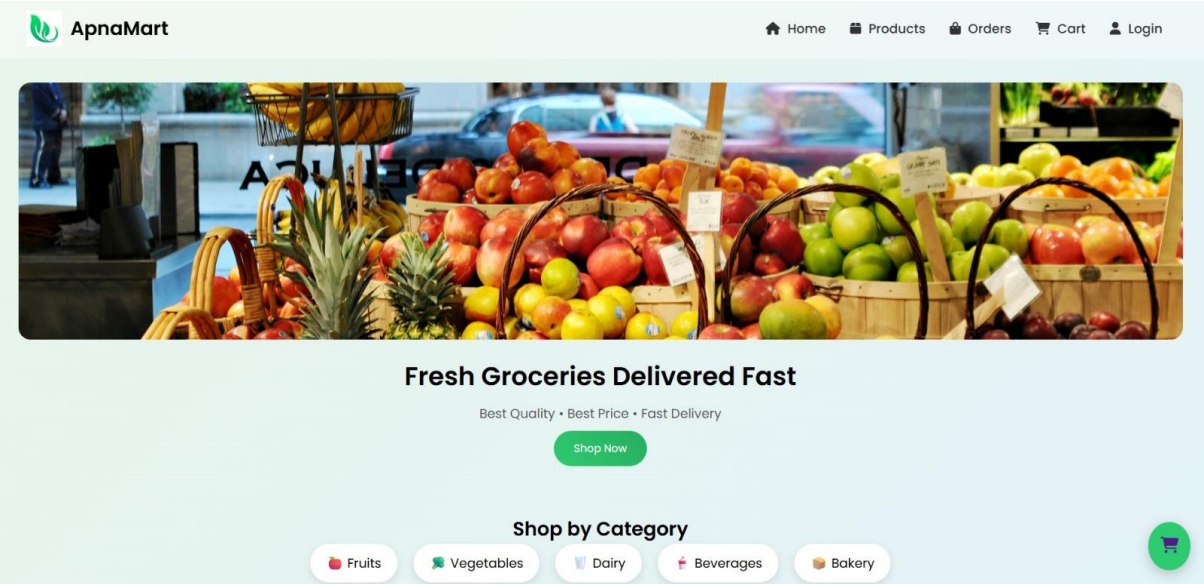
The “Files of the Project in Image” section represents the structure and organization of all files used in the development of the Online Grocery Store (ApnaMart). It shows how the project is divided into different folders such as HTML files, CSS files, JavaScript files, PHP backend files, and image assets.

This section is important because it helps the examiner understand how the project is structured and how different components are connected.

🔷 2. PROJECT FOLDER STRUCTURE


EXPLORER
OPEN EDITORS
UNTITLED (WORKSPACE)
green
css
admin_style.css
style.css
images
js
admin_script.js
script.js
uploaded_img
about.php
add_to_cart.php
admin_contacts.php
admin_header.php
admin_orders.php
admin_page.php
admin_products.php
admin_users.php
cart.php
checkout.php
config.php
contact.php
footer.php
header.php
home.php
login.php
logout.php
orders.php
README.md
receipt.php
OUTLINE
TIMELINE
0 0

Snapshot of UI



Index.html pages: This file is the main entry point of the Online Grocery Store web application. It provides the basic structure of the webpage and loads all the necessary frontend components.

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>ApnaMart - Online Grocery Store</title>

<!-- CSS -->
<link rel="stylesheet" href="css/index.css">

<!-- Font Awesome Icons -->
<link rel="stylesheet" href="https://cdnjs.cloudflare.com/ajax/libs/font-awesome/6.5.0/css/all.min.css">
```

```
</head>

<body>

<!-- ===== HEADER ===== -->
<header class="header">

    <div class="logo">
        <img src="images/leaf.jpg" alt="Logo">
        <h2>ApnaMart</h2>
    </div>

    <nav class="navbar">
        <ul>
            <li><a href="index.html"><i class="fa-solid fa-house"></i>
Home</a></li>
            <li><a href="product.html"><i class="fa-solid fa-box"></i>
Products</a></li>

            <li><a href="checkout.html"><i class="fa-solid fa-bag-shopping"></i>
Orders</a></li>
            <li><a href="cart.html"><i class="fa-solid fa-cart-shopping"></i>
Cart</a></li>
            <li><a href="login.html"><i class="fa-solid fa-user"></i>
Login</a></li>
        </ul>
    </nav>
```

```
</header>


<!-- ===== HERO ===== -->
<section class="hero">
    <img src="images/banner.jpg" alt="Banner">
    <h1>Fresh Groceries Delivered Fast</h1>
    <p>Best Quality • Best Price • Fast Delivery</p>
    <button class="btn">Shop Now</button>
</section>


<!-- ===== CATEGORIES ===== -->
<section class="categories">
    <h2>Shop by Category</h2>

    <div class="category-box">
        <div class="cat">🍎 Fruits</div>
        <div class="cat">🥦 Vegetables</div>
        <div class="cat">🥛 Dairy</div>
        <div class="cat">🥤 Beverages</div>
        <div class="cat">🍞 Bakery</div>

    </div>
</section>
```

```
<!-- ===== PRODUCTS ===== -->
<section class="products">
    <h2>Featured Products</h2>

    <div class="product-grid">

        <div class="product">
            <img src="images/apple.jpg">
            <h3>Fresh Apples</h3>
            <p>₹120 / Kg</p>
            <button>Add to Cart</button>
        </div>

        <div class="product">
            <img src="images/banana.jpg">
            <h3>Banana</h3>
            <p>₹50 / Dozen</p>
            <button>Add to Cart</button>
        </div>

        <div class="product">
            <img src="images/milk.jpg">
            <h3>Fresh Milk</h3>
            <p>₹60 / Litre</p>
            <button>Add to Cart</button>
        </div>
```

```
        <div class="product">
            <img src="images/bread.jpg">
            <h3>Brown Bread</h3>
            <p>₹40</p>
            <button>Add to Cart</button>
        </div>

    </div>
</section>



<!-- FLOAT CART -->
<div class="float-cart">
    <a href="cart.html"><i class="fa-solid fa-cart-shopping"></i></a>
</div>



<!-- FOOTER -->
<footer>
    <p>© 2026 ApnaMart Grocery Store</p>
</footer>
 <script src="js/index.js"></script>
</body>
</html>
```

admin_page.php

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>ApnaMart - Online Grocery Store</title>

<!-- CSS -->
<link rel="stylesheet" href="css/index.css">

<!-- Font Awesome Icons -->
<link rel="stylesheet" href="https://cdnjs.cloudflare.com/ajax/libs/font-
awesome/6.5.0/css/all.min.css">

</head>

<body>

<!-- ===== HEADER ===== -->
<header class="header">

    <div class="logo">
        <img src="images/leaf.jpg" alt="Logo">
        <h2>ApnaMart</h2>
    </div>

    <nav class="navbar">
```

```
        <ul>
            <li><a href="index.html"><i class="fa-solid fa-house"></i>
Home</a></li>
            <li><a href="product.html"><i class="fa-solid fa-box"></i>
Products</a></li>


            <li><a href="checkout.html"><i class="fa-solid fa-bag-shopping"></i>
Orders</a></li>
            <li><a href="cart.html"><i class="fa-solid fa-cart-shopping"></i>
Cart</a></li>
            <li><a href="login.html"><i class="fa-solid fa-user"></i>
Login</a></li>
        </ul>
    </nav>


</header>



<!-- ===== HERO ===== -->
<section class="hero">
    <img src="images/banner.jpg" alt="Banner">
    <h1>Fresh Groceries Delivered Fast</h1>
    <p>Best Quality • Best Price • Fast Delivery</p>
    <button class="btn">Shop Now</button>
</section>



<!-- ===== CATEGORIES ===== -->
<section class="categories">
```

```
    <h2>Shop by Category</h2>

    <div class="category-box">
        <div class="cat">🍎 Fruits</div>
        <div class="cat">🥦 Vegetables</div>
        <div class="cat">🥛 Dairy</div>
        <div class="cat">🥤 Beverages</div>
        <div class="cat">🍞 Bakery</div>

    </div>
</section>


<!-- ===== PRODUCTS ===== -->
<section class="products">
    <h2>Featured Products</h2>

    <div class="product-grid">

        <div class="product">
            <img src="images/apple.jpg">
            <h3>Fresh Apples</h3>
            <p>₹120 / Kg</p>
            <button>Add to Cart</button>
        </div>

        <div class="product">
```

```
            <img src="images/banana.jpg">
            <h3>Banana</h3>
            <p>₹50 / Dozen</p>
            <button>Add to Cart</button>
        </div>

        <div class="product">
            <img src="images/milk.jpg">
            <h3>Fresh Milk</h3>
            <p>₹60 / Litre</p>
            <button>Add to Cart</button>
        </div>

        <div class="product">
            <img src="images/bread.jpg">
            <h3>Brown Bread</h3>
            <p>₹40</p>
            <button>Add to Cart</button>
        </div>

    </div>
</section>


<!-- FLOAT CART -->
<div class="float-cart">
    <a href="cart.html"><i class="fa-solid fa-cart-shopping"></i></a>
```

```
</div>


<!-- FOOTER -->
<footer>
    <p>© 2026 ApnaMart Grocery Store</p>
</footer>
 <script src="js/index.js"></script>
</body>
</html>
```

Checkout.php

```
<?php

include 'config.php';

session_start();

$user_id = $_SESSION['user_id'];

if(!isset($user_id)){
   header('location:login.php');
}

if(isset($_POST['order_btn'])){
```

```
$name = mysqli_real_escape_string($conn, $_POST['name']);
$number = $_POST['number'];
$email = mysqli_real_escape_string($conn, $_POST['email']);
$method = mysqli_real_escape_string($conn, $_POST['method']);
$address = mysqli_real_escape_string($conn, 'flat no. '. $_POST['flat'].', '. $_POST['street'].', '. $_POST['city'].', '. $_POST['country'].' - '. $_POST['pin_code']);
$placed_on = date('d-M-Y');

$cart_total = 0;
$cart_products[] = '';

$cart_query = mysqli_query($conn, "SELECT * FROM `cart` WHERE user_id = '$user_id'") or die('query failed');
if(mysqli_num_rows($cart_query) > 0){
  while($cart_item = mysqli_fetch_assoc($cart_query)){
    $cart_products[] = $cart_item['name'].' ('.$cart_item['quantity'].') ';
    $sub_total = ($cart_item['price'] * $cart_item['quantity']);
    $cart_total += $sub_total;
  }
}

$total_products = implode(', ',$cart_products);

$order_query = mysqli_query($conn, "SELECT * FROM `orders` WHERE name = '$name' AND number = '$number' AND email = '$email' AND method = '$method' AND address = '$address' AND total_products = '$total_products' AND total_price = '$cart_total'") or die('query failed');
```

```
   if($cart_total == 0){
      $message[] = 'your cart is empty';
   }else{
      if(mysqli_num_rows($order_query) > 0){
         $message[] = 'order already placed!';
      }else{
         mysqli_query($conn, "INSERT INTO `orders`(user_id, name, number,
email, method, address, total_products, total_price, placed_on)
VALUES('$user_id', '$name', '$number', '$email', '$method', '$address',
'$total_products', '$cart_total', '$placed_on')") or die('query failed');
         $message[] = 'order placed successfully!';
         mysqli_query($conn, "DELETE FROM `cart` WHERE user_id =
'$user_id'") or die('query failed');
      }
   }

}

?>

<!DOCTYPE html>
<html lang="en">
<head>
   <meta charset="UTF-8">
   <meta http-equiv="X-UA-Compatible" content="IE=edge">
   <meta name="viewport" content="width=device-width, initial-scale=1.0">
   <title>checkout</title>
```

```
   <!-- font awesome cdn link  -->
   <link rel="stylesheet" href="https://cdnjs.cloudflare.com/ajax/libs/font-
awesome/6.0.0/css/all.min.css">

   <!-- custom css file link  -->
   <link rel="stylesheet" href="css/style.css">

</head>
<body>

<?php include 'header.php'; ?>

<div class="heading">
   <h3>checkout</h3>
   <p> <a href="home.php">home</a> / checkout </p>
</div>

<section class="display-order">

   <?php
      $grand_total = 0;
      $select_cart = mysqli_query($conn, "SELECT * FROM `cart` WHERE
user_id = '$user_id'") or die('query failed');
      if(mysqli_num_rows($select_cart) > 0){
         while($fetch_cart = mysqli_fetch_assoc($select_cart)){
            $total_price = ($fetch_cart['price'] * $fetch_cart['quantity']);
            $grand_total += $total_price;
   ?>
```

```
   <p> <?php echo $fetch_cart['name']; ?> <span>(<?php echo
'$'.$fetch_cart['price'].'/-'.' x '. $fetch_cart['quantity']; ?>)</span> </p>
   <?php
      }
   }else{
      echo '<p class="empty">your cart is empty</p>';
   }
   ?>
   <div class="grand-total"> grand total : <span>$<?php echo $grand_total; ?>/-
</span> </div>

</section>

<section class="checkout">

   <form action="" method="post">
      <h3>place your order</h3>
      <div class="flex">
         <div class="inputBox">
            <span>your name :</span>
            <input type="text" name="name" required placeholder="enter your
name">
         </div>
         <div class="inputBox">
            <span>your number :</span>
            <input type="number" name="number" required placeholder="enter
your number">
         </div>
         <div class="inputBox">
```

```
        <span>your email :</span>
        <input type="email" name="email" required placeholder="enter your email">
    </div>

    <div class="inputBox">
        <span>payment method :</span>
        <select name="method" id="paymentMethod" onchange="toggleQR()">
            <option value="cash on delivery">cash on delivery</option>
            <option value="UPI">UPI / QR</option>
            <option value="credit card">credit card</option>
            <option value="paypal">paypal</option>
            <option value="paytm">paytm</option>
        </select>

        <div id="qrBox" style="display:none; text-align:center; margin-top:15px;">
            <h3>Scan & Pay</h3>
            <img src="images/myqr.jpeg" width="200" alt="UPI QR Code">
            <p><b>UPI ID:</b> 844503satya@ybl</p>
        </div>
    </div>

    <div class="inputBox">
        <span>address line 01 :</span>
        <input type="number" min="0" name="flat" required placeholder="e.g. flat no.">
    </div>
```

```
        <div class="inputBox">
          <span>address line 01 :</span>
          <input type="text" name="street" required placeholder="e.g. street
name">
        </div>
        <div class="inputBox">
          <span>city :</span>
          <input type="text" name="city" required placeholder="e.g. mumbai">
        </div>
        <div class="inputBox">
          <span>state :</span>
          <input type="text" name="state" required placeholder="e.g.
maharashtra">
        </div>
        <div class="inputBox">
          <span>country :</span>
          <input type="text" name="country" required placeholder="e.g. india">
        </div>
        <div class="inputBox">
          <span>pin code :</span>
          <input type="number" min="0" name="pin_code" required
placeholder="e.g. 123456">
        </div>
      </div>
      <input type="submit" value="order now" class="btn" name="order_btn">
    </form>


</section>
```

```
<?php include 'footer.php'; ?>

<!-- custom js file link  -->
<script src="js/script.js"></script>

<script>
function toggleQR(){
   var method = document.getElementById("paymentMethod").value;
   var qr = document.getElementById("qrBox");

   if(method === "UPI"){
      qr.style.display = "block";
   }else{
      qr.style.display = "none";
   }
}
</script>

</body>
</html>
```

admin_script.js

```
let navbar = document.querySelector('.header .navbar');
let accountBox = document.querySelector('.header .account-box');

document.querySelector('#menu-btn').onclick = () =>{
```

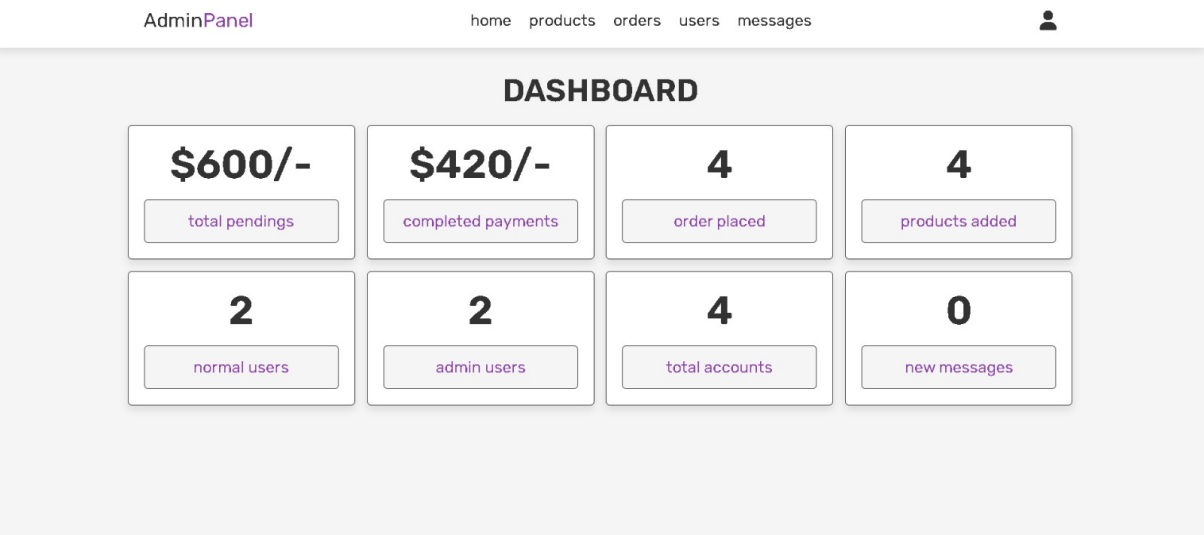
```
  navbar.classList.toggle('active');
  accountBox.classList.remove('active');
}

document.querySelector('#user-btn').onclick = () =>{
  accountBox.classList.toggle('active');
  navbar.classList.remove('active');
}

window.onscroll = () =>{
  navbar.classList.remove('active');
  accountBox.classList.remove('active');
}

document.querySelector('#close-update').onclick = () =>{
  document.querySelector('.edit-product-form').style.display = 'none';
  window.location.href = 'admin_products.php';
}
```

Script.js

```
let navbar = document.querySelector('.header .navbar');
let accountBox = document.querySelector('.header .account-box');

document.querySelector('#menu-btn').onclick = () =>{
  navbar.classList.toggle('active');
  accountBox.classList.remove('active');
}
```

```
document.querySelector('#user-btn').onclick = () =>{
  accountBox.classList.toggle('active');
  navbar.classList.remove('active');
}

window.onscroll = () =>{
  navbar.classList.remove('active');
  accountBox.classList.remove('active');
}

document.querySelector('#close-update').onclick = () =>{
  document.querySelector('.edit-product-form').style.display = 'none';
  window.location.href = 'admin_products.php';
}
```

Snapshot of Admin panel



```
<?php

include 'config.php';

session_start();

$admin_id = $_SESSION['admin_id'];

if(!isset($admin_id)){
   header('location:login.php');
}

?>

<!DOCTYPE html>
<html lang="en">
<head>
   <meta charset="UTF-8">
   <meta http-equiv="X-UA-Compatible" content="IE=edge">
   <meta name="viewport" content="width=device-width, initial-
scale=1.0">
   <title>admin panel</title>

   <!-- font awesome cdn link  -->
   <link rel="stylesheet" href="https://cdnjs.cloudflare.com/ajax/libs/font-
awesome/6.0.0/css/all.min.css">

   <!-- custom admin css file link  -->
```

```
  <link rel="stylesheet" href="css/admin_style.css">

</head>
<body>

<?php include 'admin_header.php'; ?>

<!-- admin dashboard section starts  -->

<section class="dashboard">

  <h1 class="title">dashboard</h1>

  <div class="box-container">

    <div class="box">
      <?php
        $total_pendings = 0;
        $select_pending = mysqli_query($conn, "SELECT total_price
FROM `orders` WHERE payment_status = 'pending'") or die('query
failed');
        if(mysqli_num_rows($select_pending) > 0){
          while($fetch_pendings = mysqli_fetch_assoc($select_pending)){
            $total_price = $fetch_pendings['total_price'];
            $total_pendings += $total_price;
          };
        };
      ?>
```

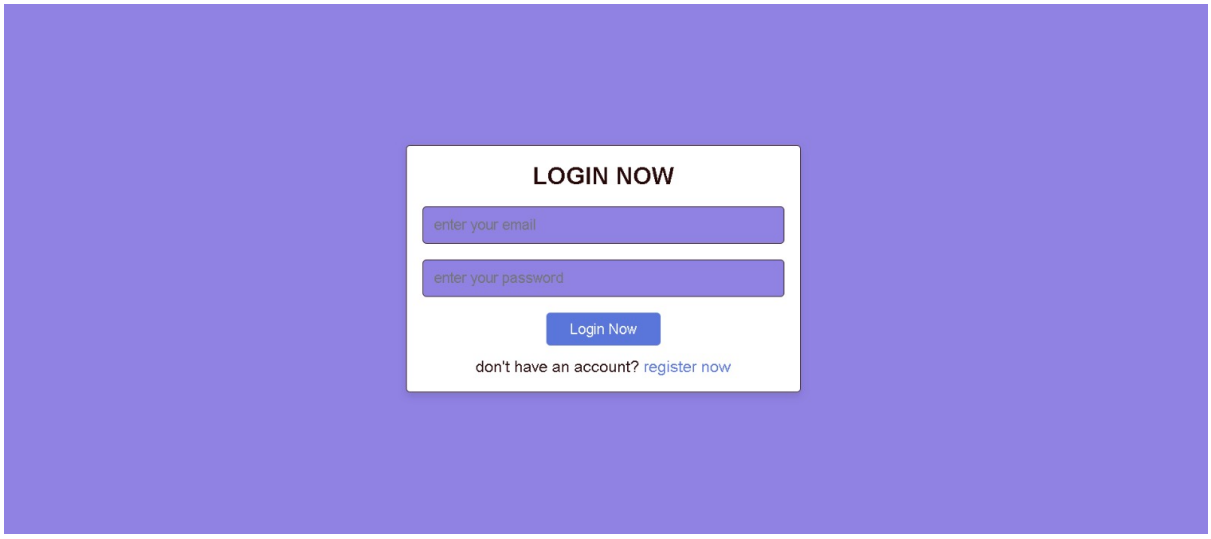
```
    <h3>$<?php echo $total_pendings; ?>/-</h3>
    <p>total pendings</p>
  </div>

  <div class="box">
    <?php
      $total_completed = 0;
      $select_completed = mysqli_query($conn, "SELECT total_price
FROM `orders` WHERE payment_status = 'completed'") or die('query
failed');
      if(mysqli_num_rows($select_completed) > 0){
        while($fetch_completed = mysqli_fetch_assoc($select_completed)){
          $total_price = $fetch_completed['total_price'];
          $total_completed += $total_price;
        };
      };
    ?>
    <h3>$<?php echo $total_completed; ?>/-</h3>
    <p>completed payments</p>
  </div>

  <div class="box">
    <?php
      $select_orders = mysqli_query($conn, "SELECT * FROM
`orders`") or die('query failed');
      $number_of_orders = mysqli_num_rows($select_orders);
    ?>
    <h3><?php echo $number_of_orders; ?></h3>
```

```
    <p>order placed</p>
  </div>

  <div class="box">
    <?php
      $select_products = mysqli_query($conn, "SELECT * FROM
`products`") or die('query failed');
      $number_of_products = mysqli_num_rows($select_products);
    ?>
    <h3><?php echo $number_of_products; ?></h3>
    <p>products added</p>
  </div>

  <div class="box">
    <?php
      $select_users = mysqli_query($conn, "SELECT * FROM `users`
WHERE user_type = 'user'") or die('query failed');
      $number_of_users = mysqli_num_rows($select_users);
    ?>
    <h3><?php echo $number_of_users; ?></h3>
    <p>normal users</p>
  </div>

  <div class="box">
    <?php
      $select_admins = mysqli_query($conn, "SELECT * FROM `users`
WHERE user_type = 'admin'") or die('query failed');
      $number_of_admins = mysqli_num_rows($select_admins);
```

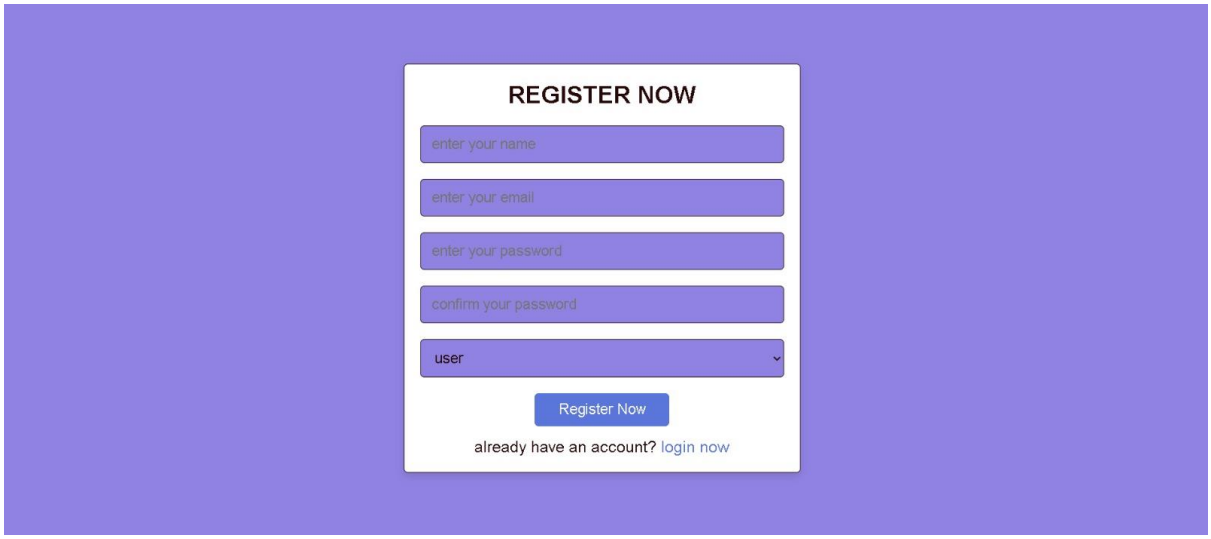
```
        ?>
        <h3><?php echo $number_of_admins; ?></h3>
        <p>admin users</p>
      </div>

      <div class="box">
        <?php
          $select_account = mysqli_query($conn, "SELECT * FROM
`users`") or die('query failed');
          $number_of_account = mysqli_num_rows($select_account);
        ?>
        <h3><?php echo $number_of_account; ?></h3>
        <p>total accounts</p>
      </div>

      <div class="box">
        <?php
          $select_messages = mysqli_query($conn, "SELECT * FROM
`message`") or die('query failed');
          $number_of_messages = mysqli_num_rows($select_messages);
        ?>
        <h3><?php echo $number_of_messages; ?></h3>
        <p>new messages</p>
      </div>

    </div>

</section>
```

```
<!-- admin dashboard section ends -->

<!-- custom admin js file link  -->
<script src="js/admin_script.js"></script>

</body>
</html>
```

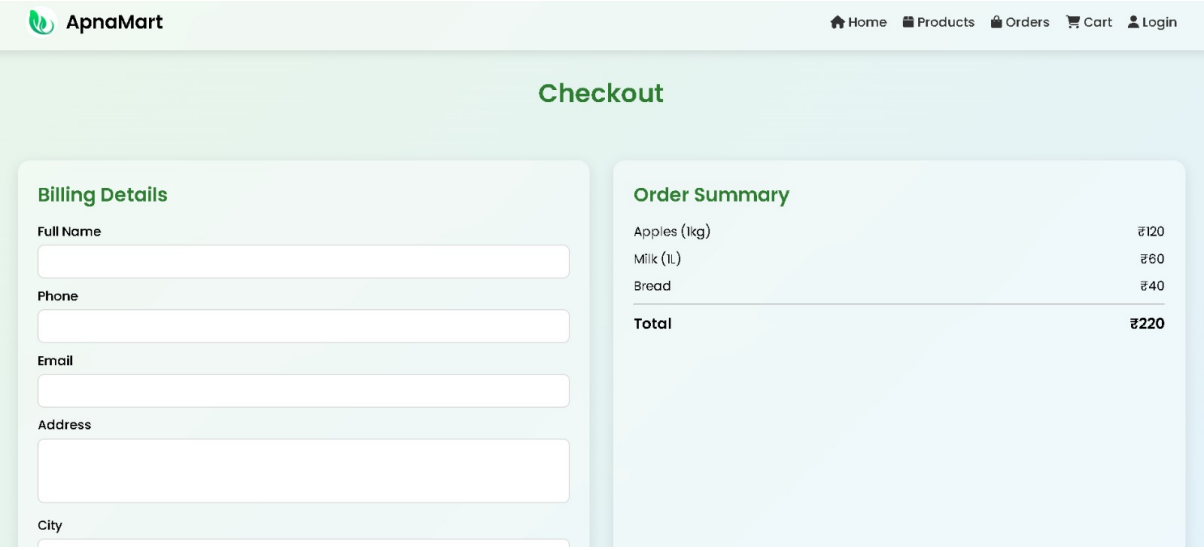
Snapshot of Login page



Snapshot of Register page



Snapshot of Checkout page



🏳 CONCLUSION

The Online Grocery Store (ApnaMart) project is a web-based application developed to make grocery shopping easy, fast, and convenient for users. It provides essential features such as user registration, login, product viewing, cart management, and order placement. The system also includes an admin module that helps in managing products, users, and customer orders efficiently.

This project demonstrates the practical implementation of web development technologies like HTML, CSS, JavaScript, PHP, and MySQL. It shows how frontend and backend systems work together to create a complete e-commerce platform. During the development of this project, important concepts such as database management, user authentication, and system design were successfully applied.

The system is user-friendly, organized, and capable of handling basic online shopping operations. Although the project has some limitations, it provides a strong foundation for future improvements such as online payment integration, mobile app development, and enhanced security features.

In conclusion, the Online Grocery Store (ApnaMart) project successfully achieves its objectives and serves as a valuable learning experience as well as a practical solution for online grocery shopping.

📚 BIBLIOGRAPHY

Books

- ***Web Technologies* – Uttam K. Roy**
- ***PHP and MySQL Web Development* – Luke Welling & Laura Thomson**
- ***Software Engineering* – Ian Sommerville**

Websites

- **www.w3schools.com**
- **www.geeksforgeeks.org**
- **developer.mozilla.org**
- **www.tutorialspoint.com**
- **www.stackoverflow.com**

Tools & Documentation

- **PHP Documentation – www.php.net**
- **MySQL Documentation – dev.mysql.com**
- **XAMPP – www.apachefriends.org**

Others

- **YouTube Tutorials (for project guidance)**
- **Online coding resources and notes**